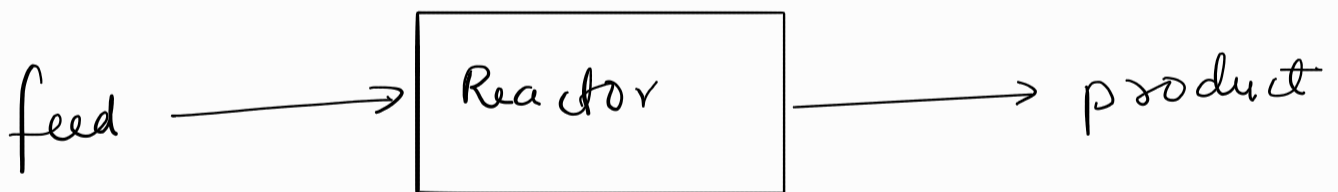
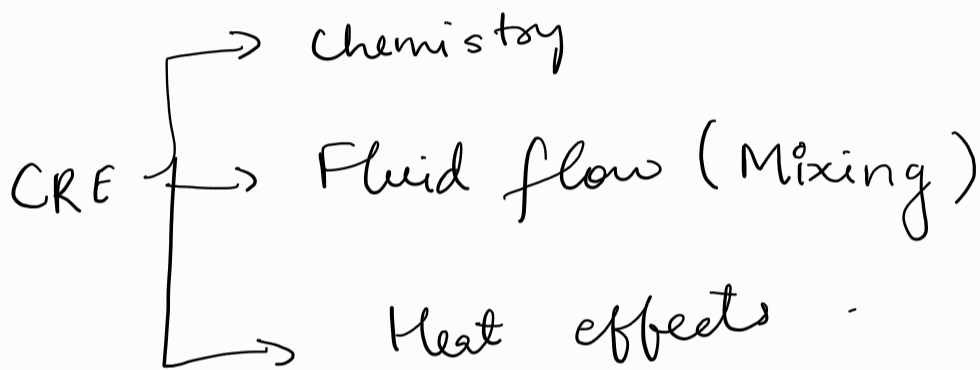
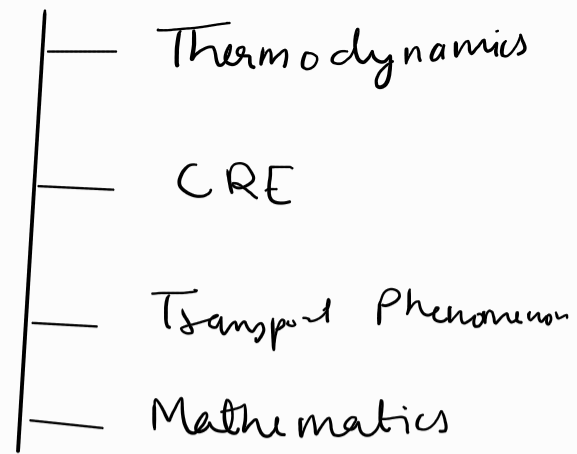


C.R.E. II



$$\text{product} = f \left(\overset{\text{rate}}{\text{feed}}, \overset{\text{pressure}}{\text{conc}^n}, \text{temp}^{\leftarrow}, \text{volume} \right)$$

↳ Kinetics, flow pattern

↓
get of reactants



$$-\gamma_A = f(C_A, T)$$

Assumption

- ① mixing at molecular level.
- ② Flow pattern.

we need high n.o.r $\Rightarrow$ less volume.

$$-\gamma_A = f(\text{collision frequency})$$

$\swarrow$ by
 ① inc concn
 (partial pressures)
 of gas

$\begin{cases} \rightarrow \text{frequency} \\ \rightarrow \text{orientation of molecules} \\ \rightarrow \text{Energy.} \end{cases}$

$$-\gamma_A = f(T) \rightarrow \text{Arrhenius theory}$$

$$k = k_0 e^{-E_A/RT}$$

$\downarrow$
 always dominant.

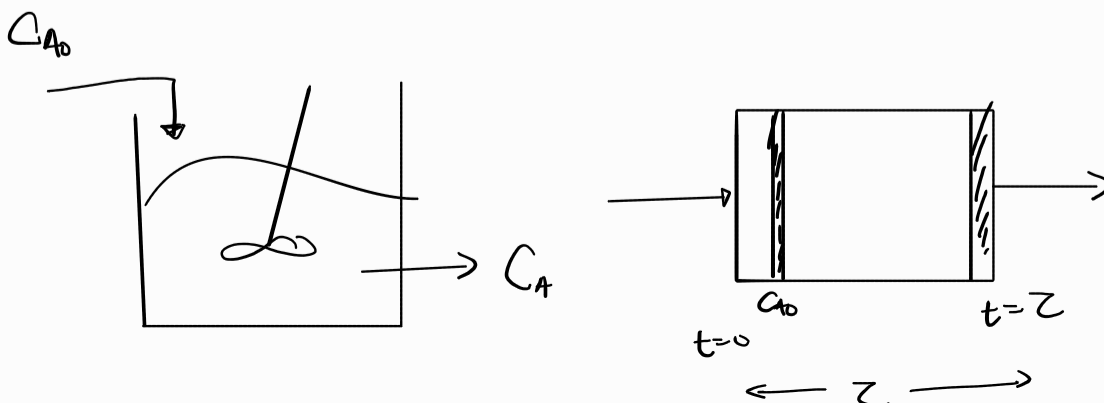
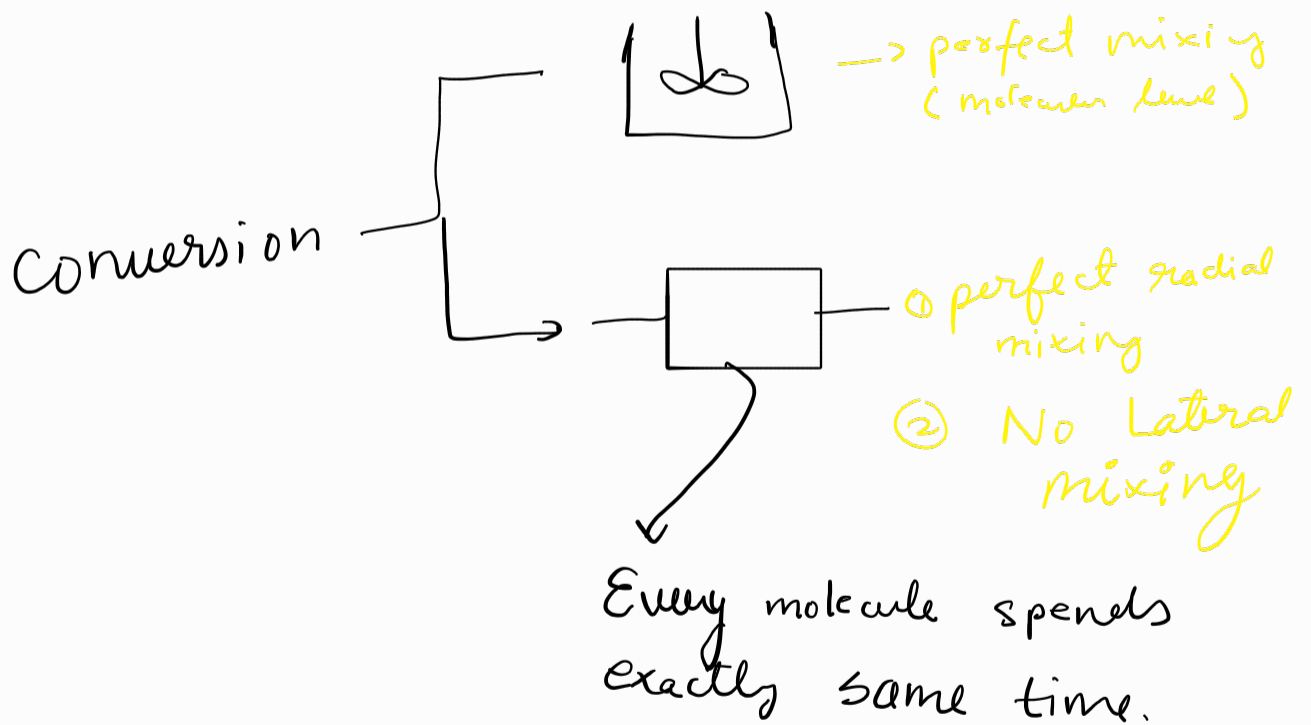
$$-r_A = k C_A^n$$

order (found through experiment)

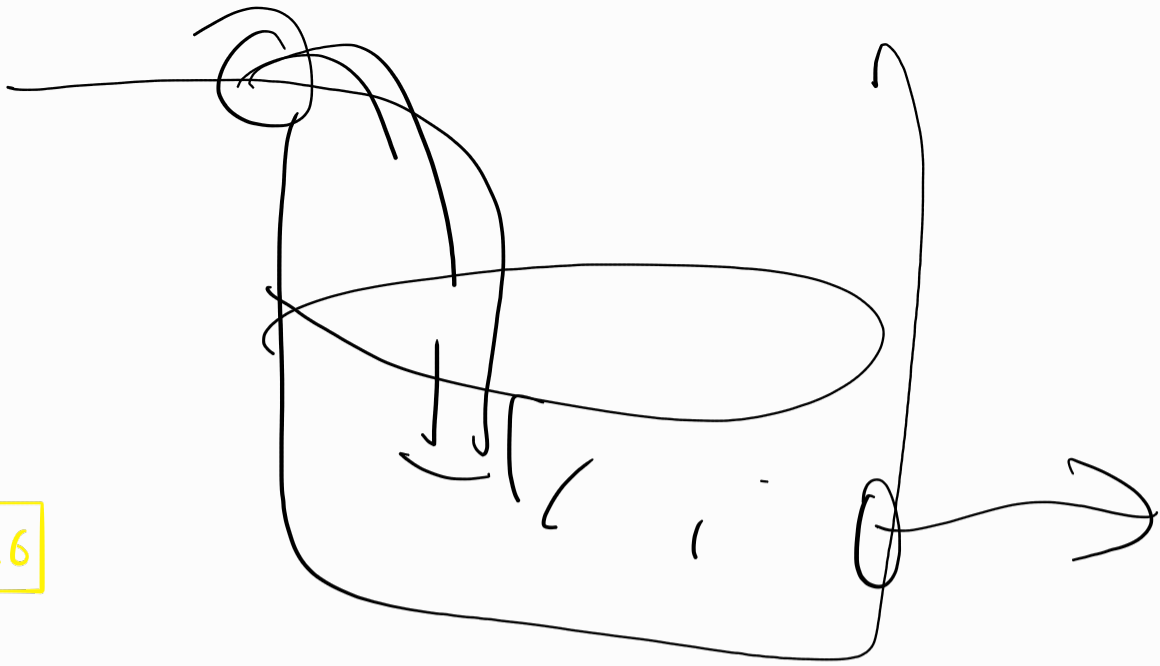
$$n < 3$$

↳ no. of molecules colliding

We can't calculate the conversion if we have any other reactor other than CSTR & PFR.



29/07/26



Why CSTR has low
Conversion than PFR

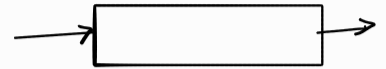
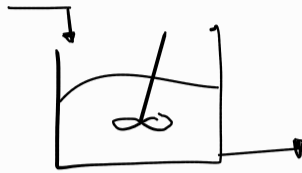
- ⊗ When PFR behaves as CSTR & vice-versa
- ⊗ What will happen if we add recycle to CSTR?
- ① Calculate 'V' for given 'x'
- ② Calculate 'x' for given 'V'

Ideal Continuous Reactors

Mixed flow
(CSTR)

plug flow reactor

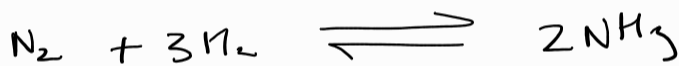
Representation



performance
eqn

$$\frac{Z}{C_{A0}} = \frac{V}{F_{A0}} = \frac{x_A}{-r_A}$$

$$\frac{Z}{C_{A0}} = \frac{V}{F_{A0}} = \int \frac{dx}{-r_A}$$



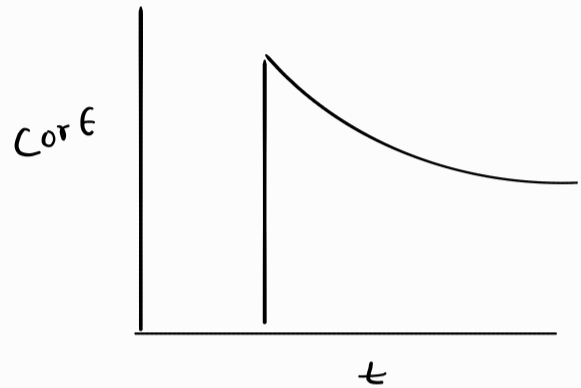
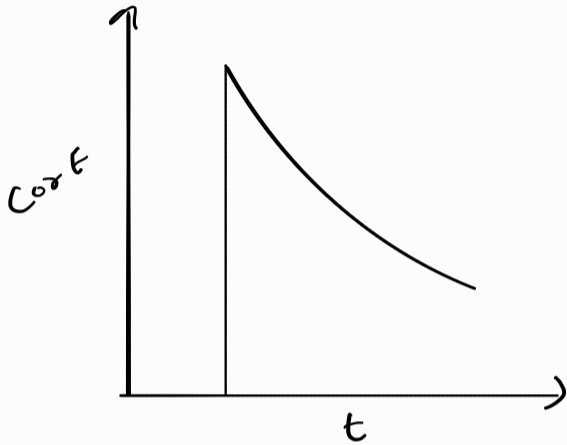
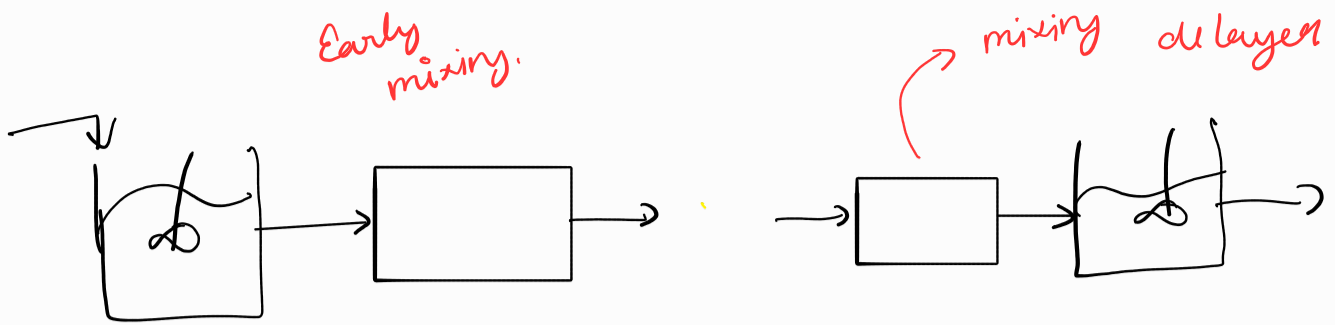
reaction equilibrium

equilibrium v/s steady state

Note: whenever we have a variation of something in space we end up with integration.

→ For CSTR molecules of different residence times get mixed with each other.

→ For PFR they only mix with their own batch having same residence time.



⊗ Mixing doesn't influence $n=1$

⊗ Late mixing gives greater conversion for $(n>1)$

3/08/26 → in slide { Started
Heterogeneous
rxn }

4/08/26

Heterogeneous reaction

Catalytic

- Hydro cracking
- Cat. cracking, reforming
- Hydro / Dehydrogenation
- Desulfonation
- Alkylation

G/S

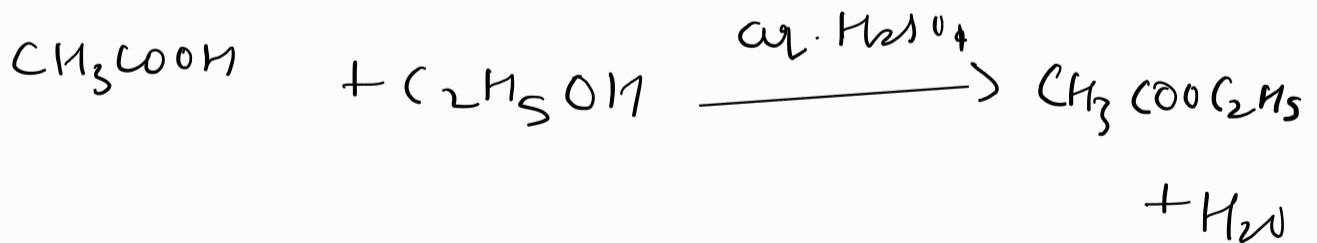
G/L/S

L/S

Non-Catalytic

- Combustion
- Gasification

Catalytic - homogeneous rxn



- ① It alters the rate but not thermodynamics.
- activation energy changes.
(doesn't alter the eq^m)
- ↓, ΔG
← ΔH will
be same
for both.
- ② It is not consumed but they deactivate

- (3) They provide high surface area
 $\sim 10^2 - 10^3 \text{ m}^2/\text{gm}$

- ④ Supported -
- mechanical strength
favourable sites.
thermal stability
- support
- promoter
inhibitor
- SiO_2
 Al_2O_3



Th Miller

द्वि-
को वारे में
वात सुस ।

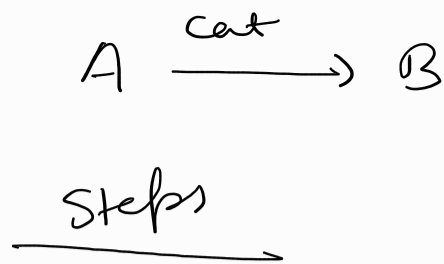
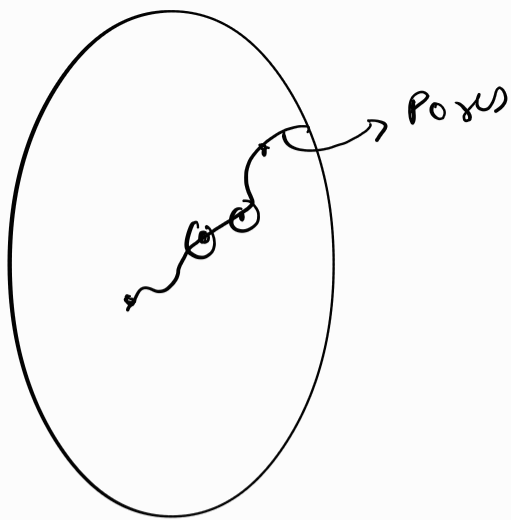
→ Unsupported — Fe for NH_3 production

$$-r_v = \frac{1}{V} \frac{dN}{dt} \left[\frac{\text{moles}}{\text{m}^2 \text{-reactor-s}} \right]$$

$$-r_w = \frac{1}{W} \frac{dN}{dt} \left[\frac{\text{moles}}{\text{kg. cat. - s}} \right]$$

$$-r_s = \frac{1}{S} \frac{dN}{dt} \left[\frac{\text{moles}}{\text{m}^2 \text{-cat-s}} \right]$$

$$(-r_v) V = (-r_w) W = (-r_s) S$$



- ① Mass transfer $\left(\begin{array}{l} \text{of 'A' to surface of} \\ \text{catalyst } [C_A, k] \end{array} \right. \xrightarrow{\text{MTC}}$
- ② Diffusion of 'A' to active site.

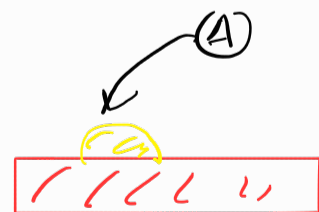
Catalyst size

$50 \mu\text{m} - 3600 \mu\text{m}$

pore size

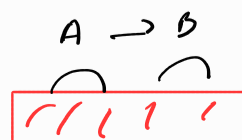
————→ A°

- ③ Adsorption of 'A'

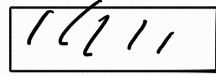


- ④ Surface reaction of A to B ($A \rightarrow B$)

adsorption & reaction can happen at different sites.



(5) Desorption of B (B) $\rightarrow$



(6) Diffusion of B to catalyst surface

(7) Mass transfer of 'B' to bulk.

(8) How to calculate M.T.C ?

(9) How to control M.T.C ?

Shrunk number

(10) adsorption ... $\rightarrow$ surface chemistry.

Catalyst Shapes

Spherical — low manufacturing cost, relatively high pressure drop, large diffusion length

Application — Hydrodesulfurization (HDS)
Methanation

is regular
↓
close packing
↳ less voids
→ low m.c., high p.d., low surface area.

Application — Ammonia, formaldehyde

Pellet — regular shape, good strength, low surface area.
application — CO shift, Hydrogenation



Ring



Miniball



wagon wheel.

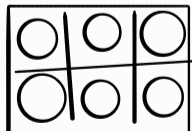
Extruded cylinder.

— low pressure drop, poor strength

Application — HDS

↳ tensile & compression

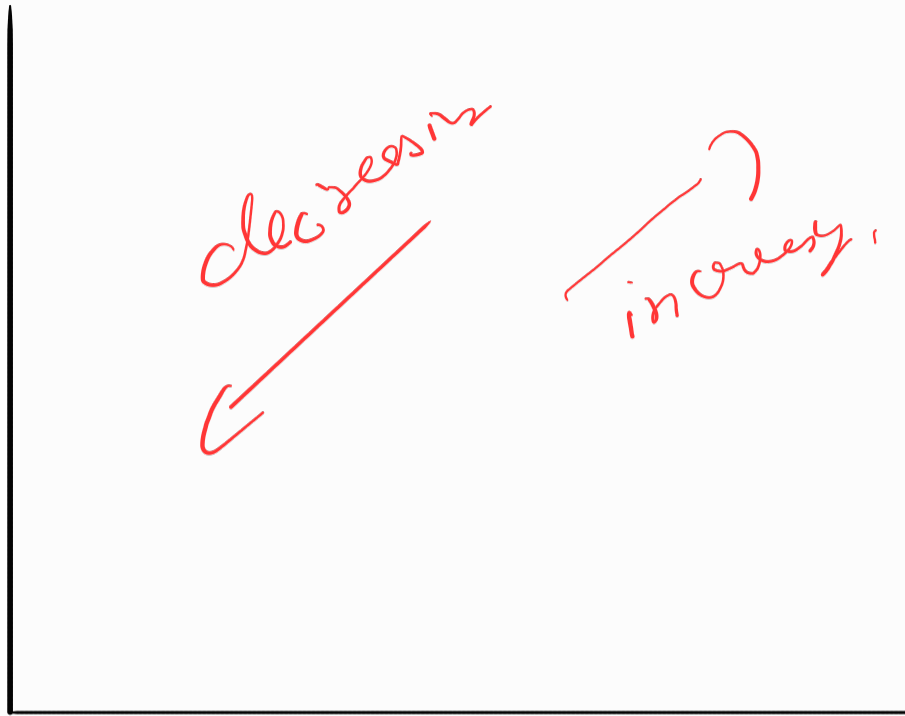
Monolith



→ low p.d., small diffusion length,
poor radial heat transfer.

Application — Exhaust Gas cleaning.

Effect of Catalyst shape - surface area



$$\frac{\Delta P}{L} = \frac{150 \mu V (1-E)^2}{d_p^2 E^3} + \frac{1.78 f v^2}{d_p E^2} (1-E)$$

less rate $\rightarrow$ higher amount of catalyst



that's why we need a minimum rate otherwise the cost will be high.

high p.d $\rightarrow$ bigger pump
will be needed
 $\searrow$
small catalyst
particle
(less voidage)
additional cost.

Strength — tensile & compression

Fixed Bed $\rightarrow$ solid are in rest
fluidized bed $\rightarrow$ in movement.

Catalyst Types and Preparation

(Bulk / Unsupported)

Precipitation / Gel

...

Decantation / Filtration

Washing

Drying

Crushing

Calcination

heat in absence of O_2

Impregnation

Activation

generic method
for catalyst
preparation
but may not be
true.

Metals (Fe, Cu, Pt, Pd, Rh)

Semicon (NiO, ZnO)

K₂O Cl

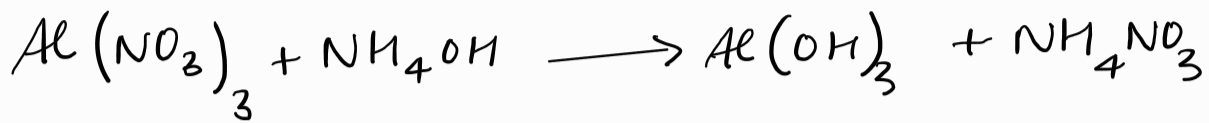


Promoters / Inhibitors

SiC
Silicon
carbide

Insulators (Al_2O_3 , SiO_2)

Example - precipitation



Solgel method

- ① Precipitation of liquid suspension by hydrolysis of precursor salt.
- ② Condensation leads to gel formation

Supported Catalyst

Dry, and wet

Impregnation

Decantation / Filtration

Washing

Drying

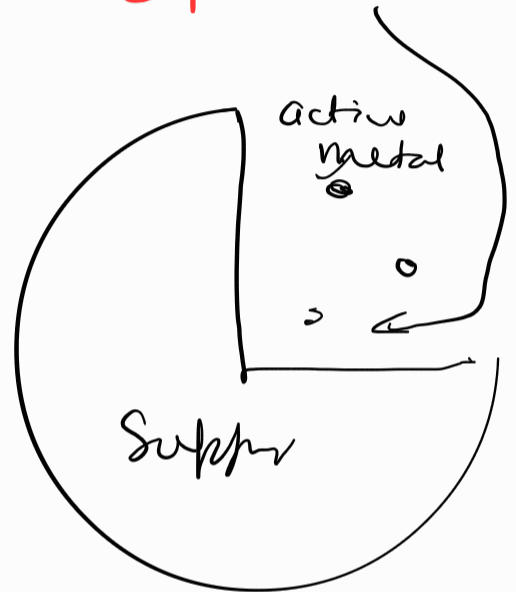
Crushing

Calcination

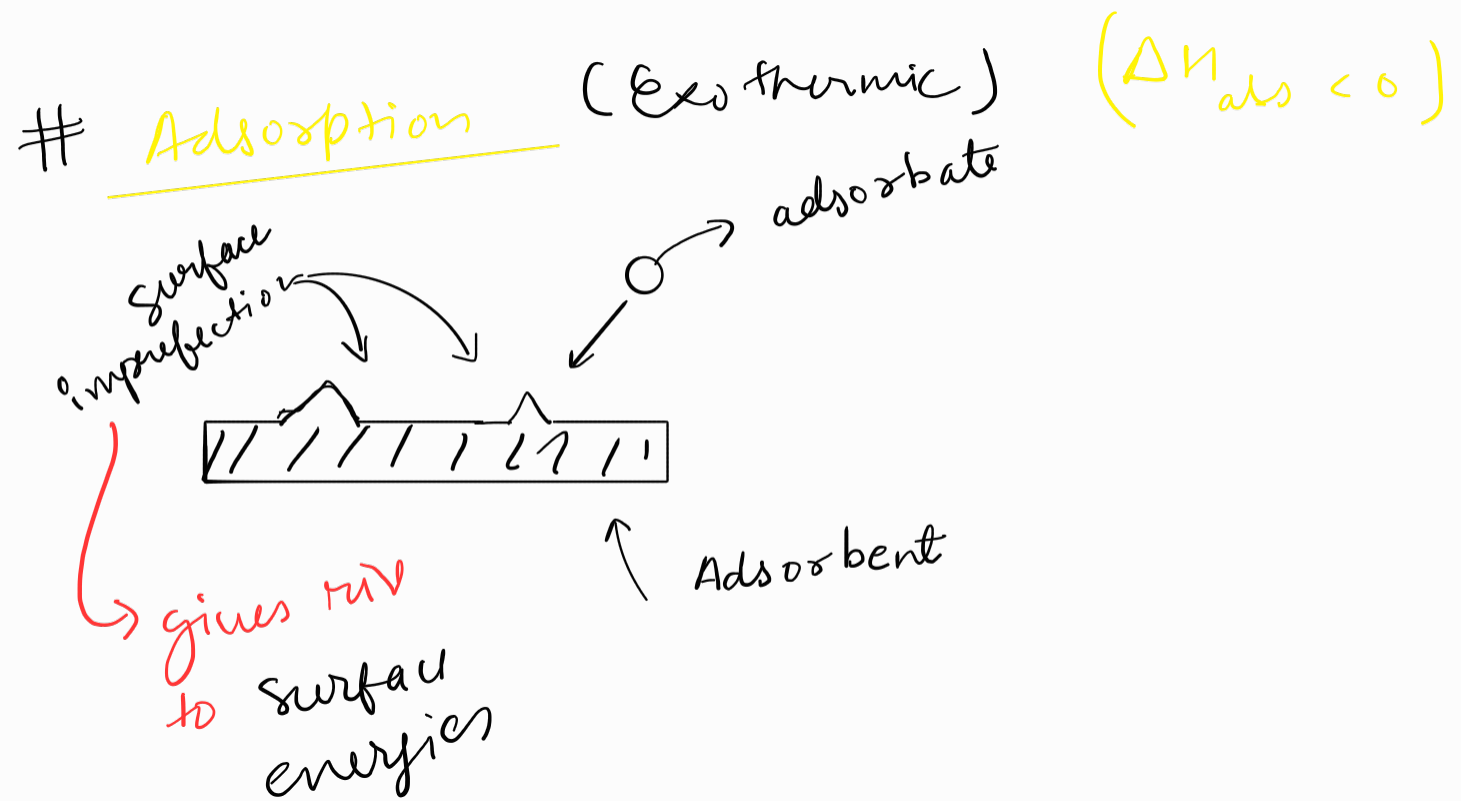
Impregnation

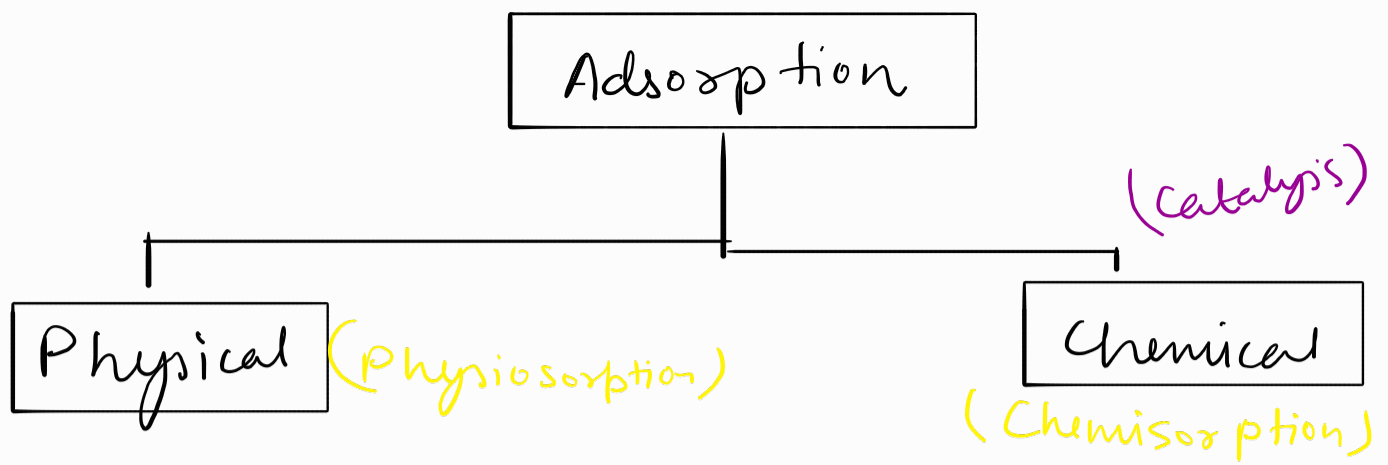
Activation

Diffusion
+ capillary flow



Dry (exact
precursor
salt)





① Van der Waal's

② Any surface

③ Low ΔH_{ad}

(0.5 - 5 kcal/g-mol)

$\sim \Delta H_{\text{condensation}}$

④ Low 'T'



⑤ Reversible

① Chemical bond

② Chemically active

③ High ΔH_{ads}

(5 - 100 kcal/g-mol)

$\sim \Delta H_{\text{bond energy}}$

④ High Temperature.

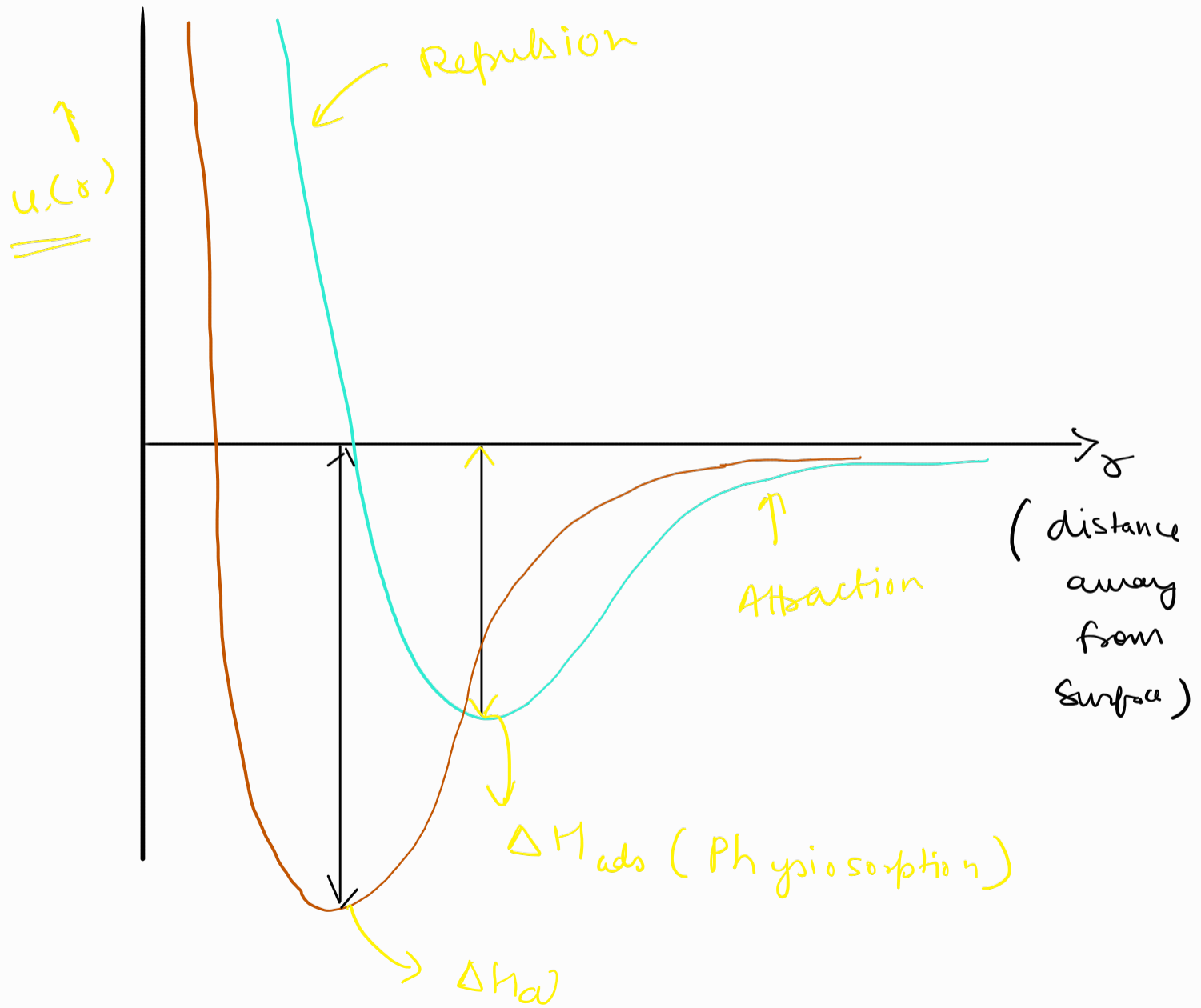


⑤ Irreversible

⑥ single layer
after that
physisorption

we use chemisorption in catalysis

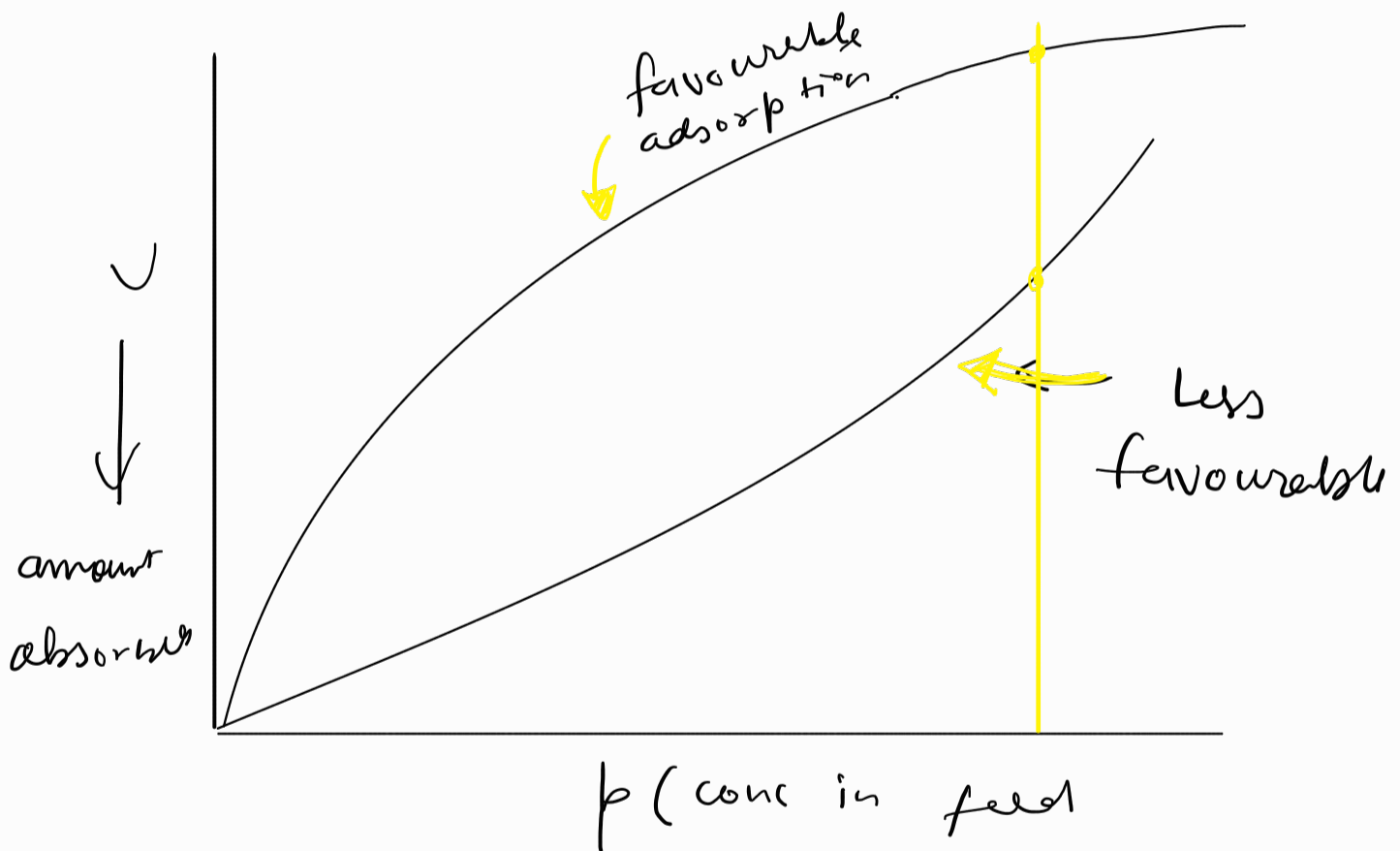
Intermolecular Potential Energy (diagram) (Pair potential)



Adsorption Isotherm

↳ Plot of quantity of adsorbate on adsorbent at equilibrium, at const Temp

$$(\delta_{ads} = \delta_{des})$$



Langmuir isotherm

Assumptions:

- ① Adsorbent surface is uniform.
- ② Adsorbed molecules don't interact - no interaction



③ ΔH_{ads} is constant and independent of coverage (fraction of occupied sites)

④ less than a monolayer.

5 sites avail

1 molec. trying to adsorb

1 site available
1 molec. trying

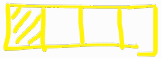
If 5 sites are available &
1 adsorbate molec. is coming &

If 1 site is available &
1 molec. is coming

Same ΔH_{ads} will be released.

$n_{adsorption} \propto \bar{F} \pi_{collision} \propto k p$
 $\downarrow$ $\downarrow$ $\downarrow$
 Fraction adsorbed adsorption constant partial pressure

$\propto (1 - \theta)$

 $\theta = 1/4$

fraction of site occupied.

$$n_{ads} = k p (1 - \theta)$$

$$n_{desorption} = k' \theta$$

⇒ At equilibrium

$$r_{ads} = r_{deso}$$

$$\Rightarrow k p (1 - \theta) = k' \theta$$

$$\Rightarrow k p - k p \theta = k' \theta$$

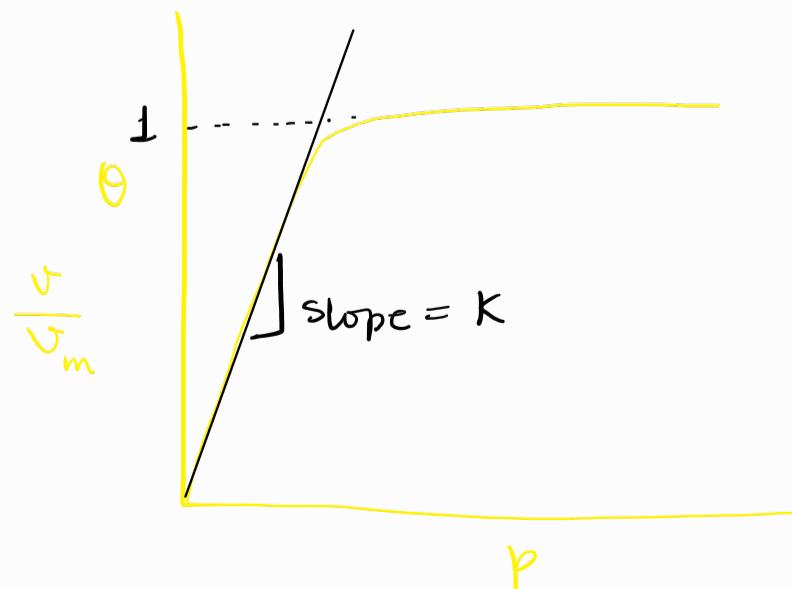
$$\Rightarrow \theta (k' + k p) = k p$$

$$\Rightarrow \theta = \frac{k p}{(k' + k p)}$$

$$K = \frac{k}{k'} = \text{Adsorption equilibrium constant.}$$

$$\theta = \frac{K \cdot p}{(1 + K p)} = \frac{V}{V_m} = \frac{\text{volume of surface adsorbed}}{\text{total available volume}}$$

$\downarrow$
monolayer volume



For large p $\theta = 1$
(partial pressure)

$$\begin{aligned} K p < 1 & \quad \theta = K p \\ K p > 1 & \quad \theta = 1 \end{aligned}$$

10/8/26

Adsorption isotherm

1) Langmuir isotherm

$$\frac{V}{V_m} = \theta = \frac{Kp}{1 + Kp}$$

Annotations:

- V_m is gas adsorbed. (points to V)
- V_m is monolayer volume. (points to V_m)
- θ is coverage. (points to θ)
- K is adsorption eqm const. (points to K)
- p is partial pressure. (points to p)

2) Temkin isotherm

$$\Delta H_{ads} \propto \frac{1}{\theta}$$

$$\theta = k_1 \ln(k_2 p)$$

$\{ k_1, k_2 \text{ constant at } (T) \}$

(3) Freundlich isotherm

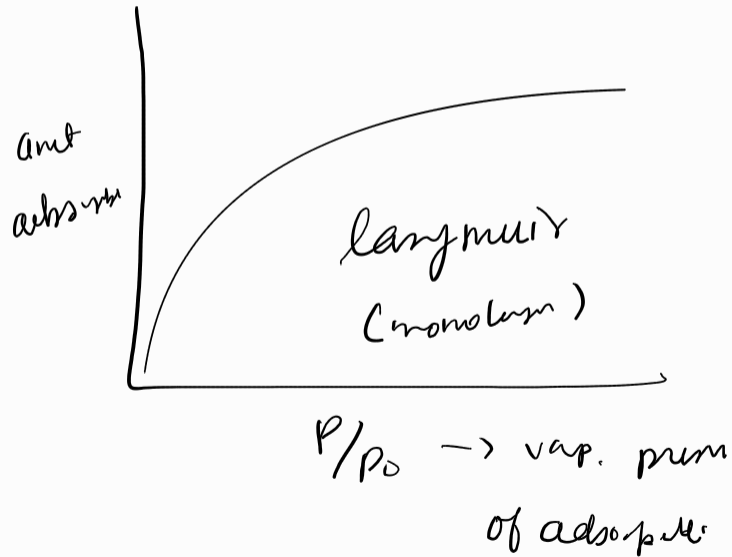
$$\Delta H_{ads}(\theta) = \Delta H_0 \ln \theta$$

$$\theta = C(p)^{1/n} \quad n > 1$$

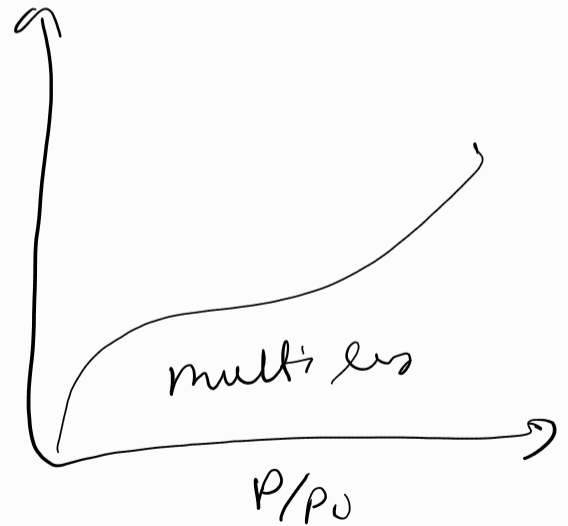
classification of isotherm

(IUPAC Classification)

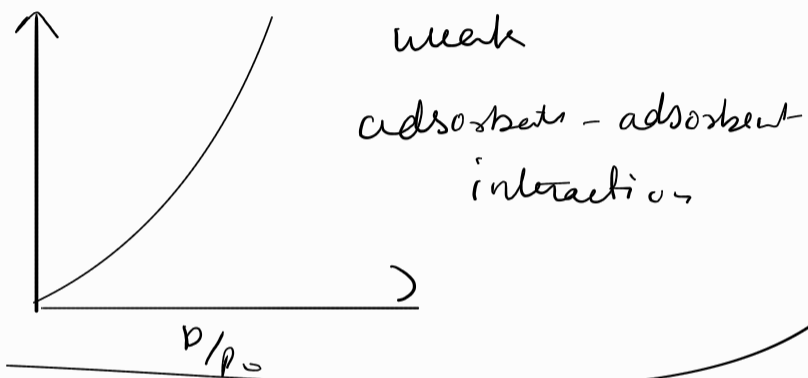
1) Type-1



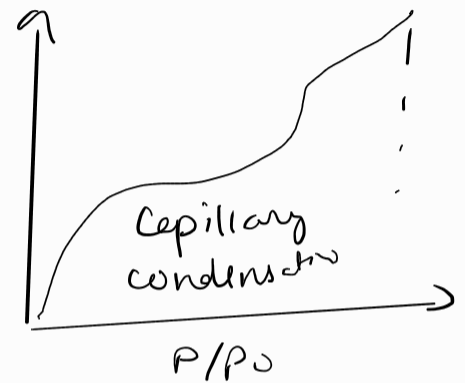
2) Type 2



(3) Type III



(4) Type IV



(5) Type-S



weak adsorption
+
capillary condensation

① Physisorption = surface area + pore volume.

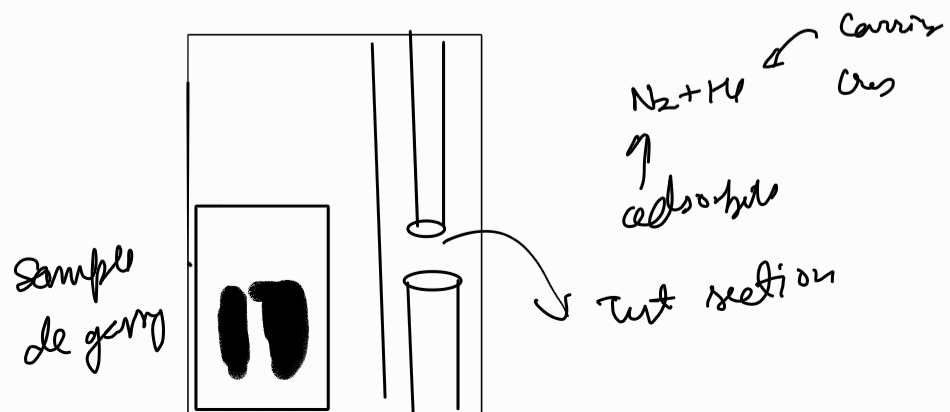
② chemisorption $\rightarrow$ active sites.

Types of pores (IUPAC)

① Micropore $d_p < 2 \text{ nm}$

② Mesopore $2 \text{ nm} < d_p < 50 \text{ nm}$

③ Macropore $d_p > 50 \text{ nm}$



Brunauer - Emmett - Teller (BET) Methode

$$\frac{V}{V_m} = \frac{kP}{1+kP}$$

$$\frac{P}{V(P_0 - P)} = \frac{1}{\underbrace{V_m P}_C} + \frac{C-1}{C V_m} \left(\underbrace{\frac{P}{P_0}}_x \right)$$

→ BET adsorption isotherm

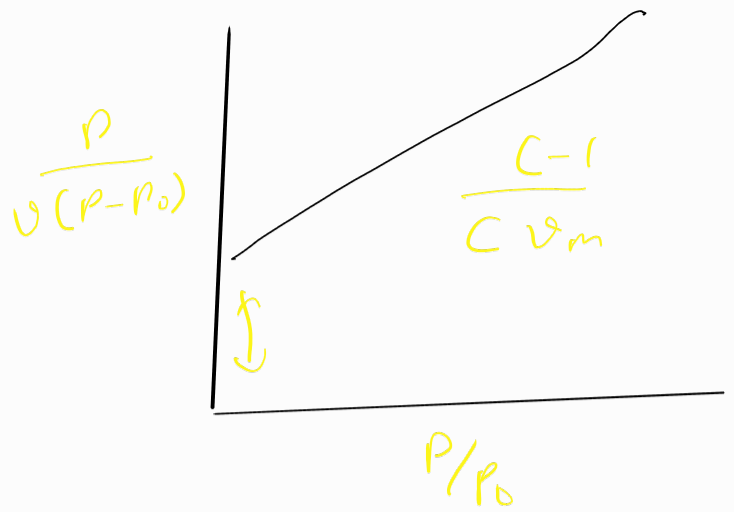
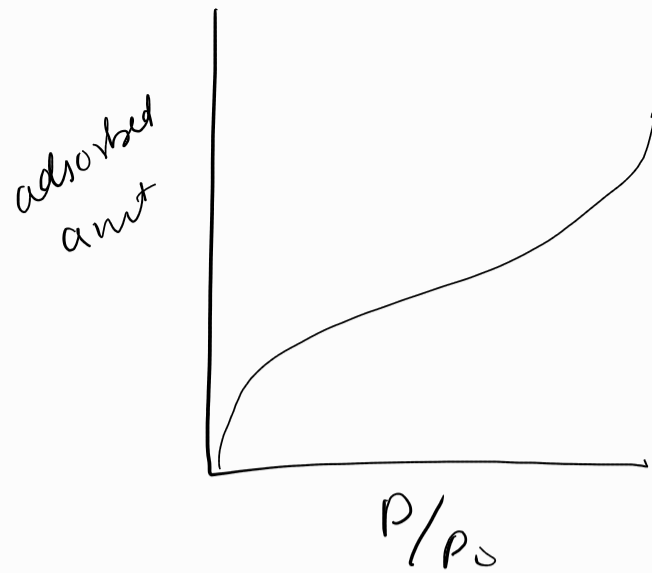
P_0 : vapour pressure of adsorbate

C : BET constant

$$C = \exp \left[\frac{E_1 - E_L}{RT} \right]$$

E_1 : heat of adsorption

E_L : heat of vaporization



$$I = \frac{1}{V_{mp}}$$

$$S = \frac{C-1}{C V_m}$$

$$V_m = \frac{1}{I \cdot S}$$

$$V_m = \frac{N_A V_m}{V_{stp}}$$

Let (α) be area of one molecule

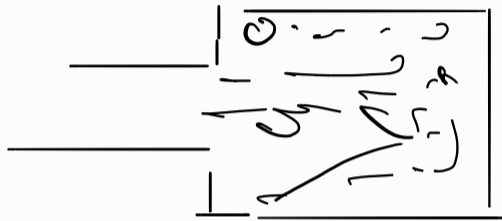
$$S = \frac{N_A V_m}{V_{stp}} \cdot \alpha$$

↪ specific surface area.

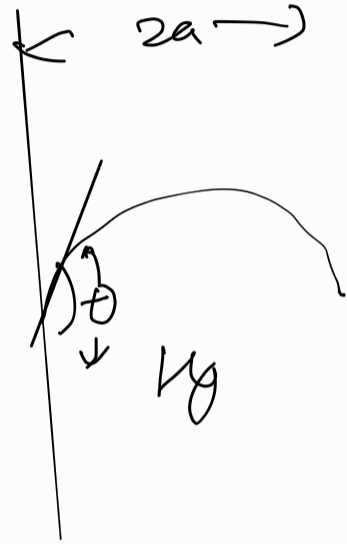
for specific surface area divide by
(m) man.

Pore Volume - Mercury Intrusion method

$$a = \frac{-2\sigma \cos\theta}{P}$$



mercury



$$a = \frac{-2\sigma \cos\theta}{P}$$

Turnover frequency

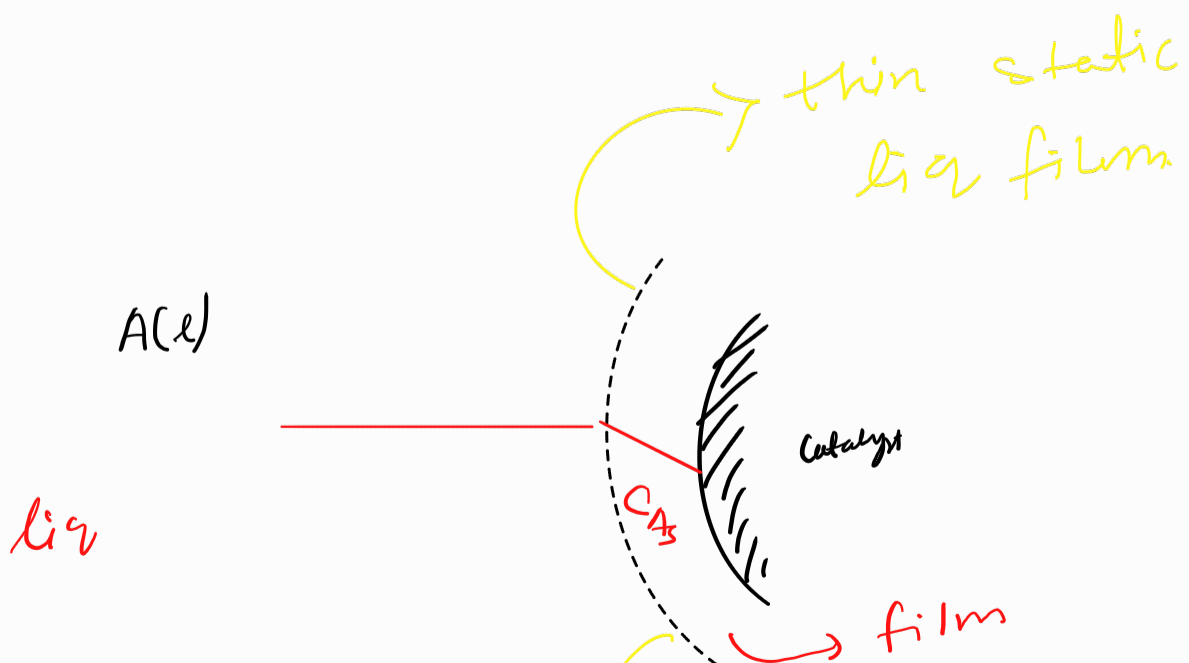
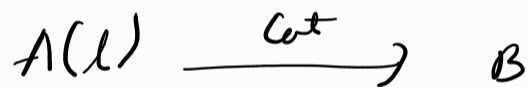
$$= \frac{\text{moles of reactant converted / time}}{\text{moles of active sites}}$$

(s^{-1})
OR

(mol / active site.s) $\checkmark$ (10^{-4} to 10^4)

Chemisorption

$$\text{Dispersion} = \frac{\text{Number of surface metal atoms}}{\text{Total no. of metal atoms.}} [\%]$$



entire resistance of liquid phase lies inside this film

① Transfer of 'A' to surface of catalyst

$$-r_A = k_g a (C_{Ab} - C_{As})$$

rate of transport of A due to M.T. $\rightarrow$ interstitial area $\rightarrow$ MTC

$$N_A = -D \frac{dc_A}{dz}$$

unknown

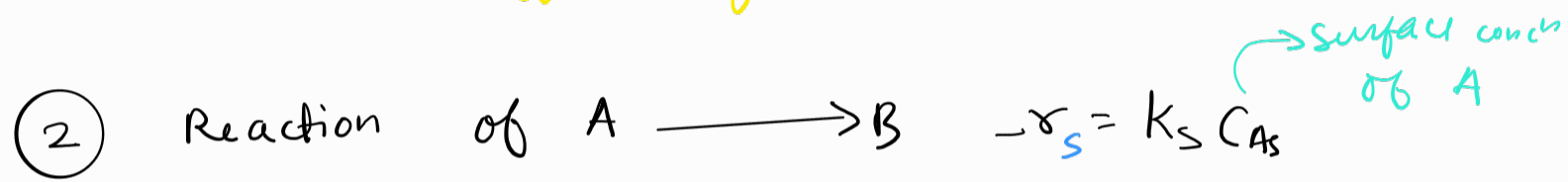
$$N_A = k (C_{A1} - C_{A2})$$

Assumption

① Static film (thickness not known)

② Mass transfer is through diffusion

③ Entire bulk resistance is equal to resistance offered by film.



at steady state

$$-r_A = -r_s$$

$$\Rightarrow k_g a (C_{Ab} - C_{As}) = k_s C_{As}$$

What is the rate expression for heterogeneous (liq-solid) reaction?

Can't mean $-r_s = k_s C_A$

$\rightarrow$ can't mean on surface of catalyst.

$$C_{Ab} = \text{known}$$

$$\Rightarrow k_{Ag} a (C_{Ab} - C_{As}) = k_s C_{As}$$

$$\Rightarrow k_{Ag} a C_{Ab} - k_{Ag} a C_{As} = k_s C_{As}$$

$$C_{As} (k_s + k_{Ag} a) = k_{Ag} a C_{Ab}$$

$$-r_s = \frac{k_s k_{Ag} a C_{Ab}}{k_s + k_{Ag} a}$$

$$-r_s = \frac{C_{Ab}}{\frac{1}{k_s} + \frac{1}{k_{Ag} a}}$$

Kinetic
resistance

Mass transfer
resistance

Extrinsic / Global
/ observed rate
of reaction

$$-r_s = k_s C_{Ab} \longrightarrow \text{intrinsic}$$

- ① Increase C_{As}
 - ② Increase a
 - ③ Increase k_{Ay} (Sherwood no.)
 - ④ Increase k_s (high temp)
- high velocity

Case I

$$k_{Ay} a \gg k_s$$

all the resistance is offered by intrinsic nature.
(Reaction / Kinetic controlled)

$$\frac{1}{k_s} \gg \frac{1}{k_{Ay} a}$$

Case II

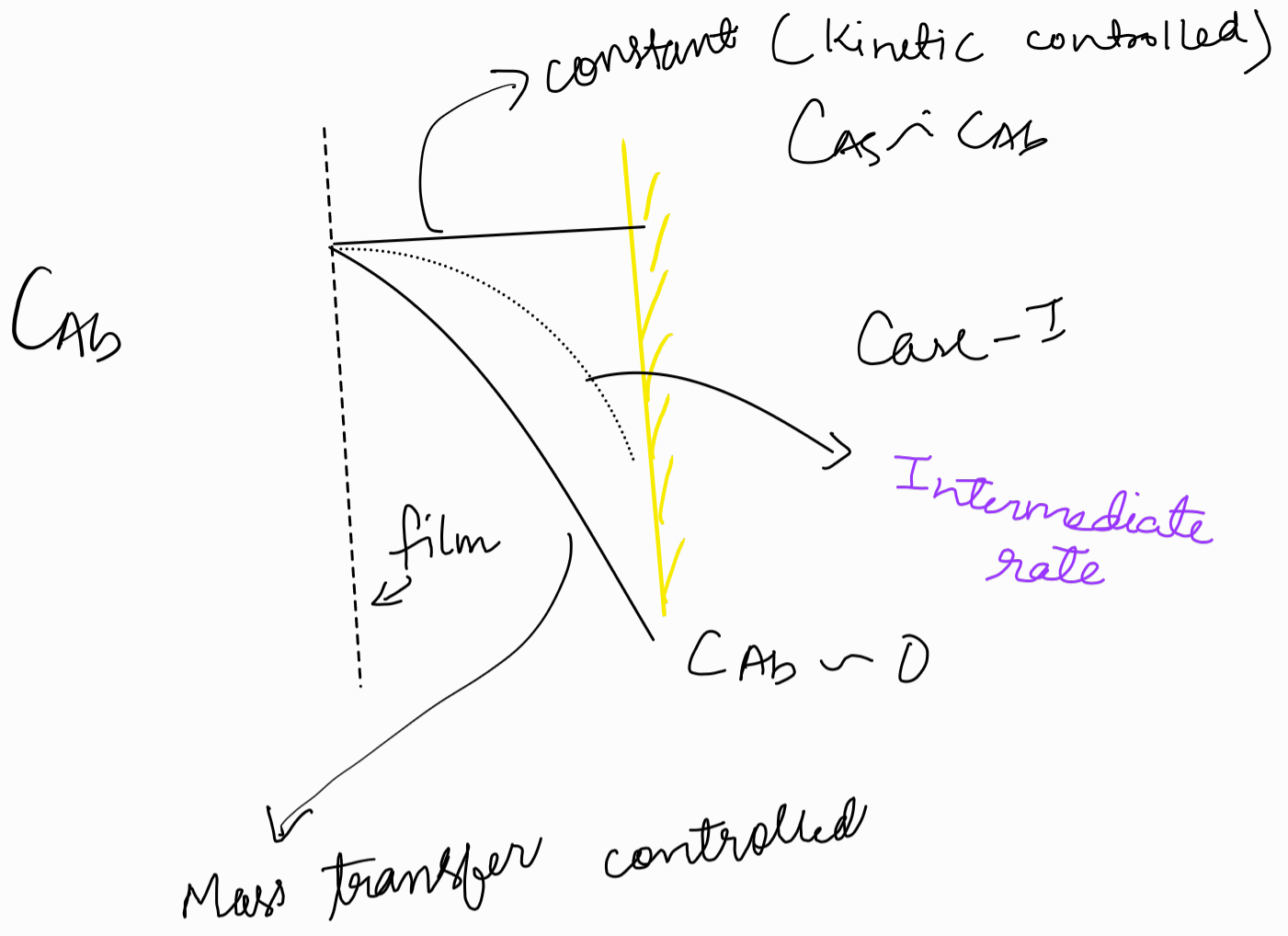
$$k_{Ay} a \ll k_s$$

Mass transfer controlled

$$\frac{1}{k_{Ay} a} \gg \frac{1}{k_s}$$

$$-r_s = k_{Ay} a C_A$$

$$[C_A < C_{As}]$$



$\gamma_s = \text{very very small no.}$
 (Vatschal's di h)

$$f(\gamma_A)_D = k_1 C_{Ab}$$

Next Tuesday Quiz-1

25th August

↓
until 14th August (T-1)

$$-r_A = \frac{C_{A_{\text{outer}}}}{\frac{1}{k_S} + \frac{1}{k_g a}}$$

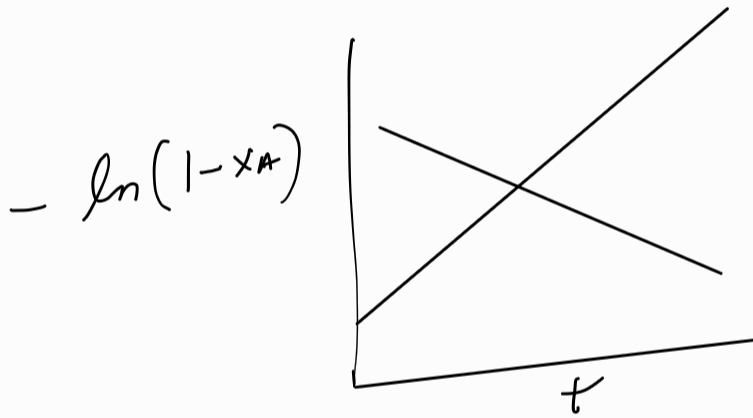
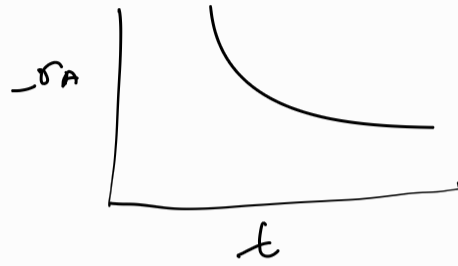
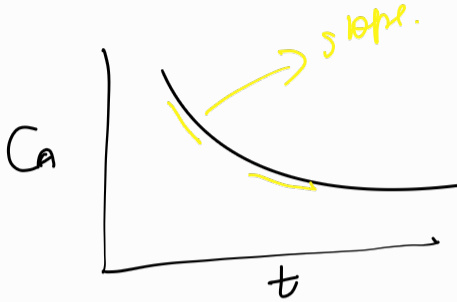
$$-r_A = f(k_S, k_g, a, C_A)$$

$$\frac{V}{F_{A0}} = \frac{X}{-r_A}$$

$$\frac{V}{F_{A0}} = \int \frac{dX}{-r_A}$$

if not correct

① Perform experiment.



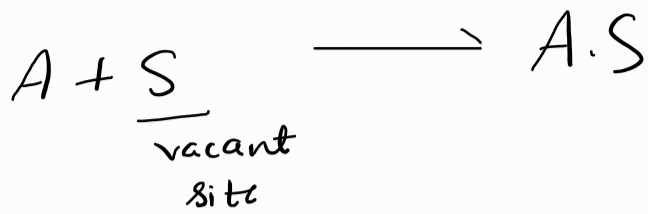
② Assume a rate law [rate limiting step]

③ Deduce parameters

if correct

④ Reactor design

Adsorption (chemisorption) $\rightarrow$ adsorbed A {notation for gln}



$p_A \rightarrow$ partial pressure.

(1) Reversible

$C_v \rightarrow$ vacant site

(2) Elementary

$$\left[\frac{\text{mol}}{\text{g. cat}} \right]$$

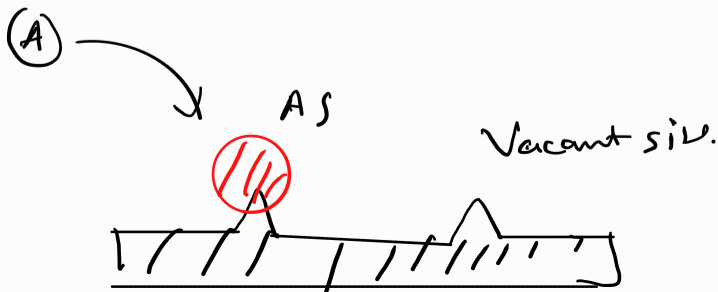
$C_t \rightarrow$ Total site $\left[\frac{\text{mol}}{\text{g. cat}} \right]$

$C_t, C_{A.S}$

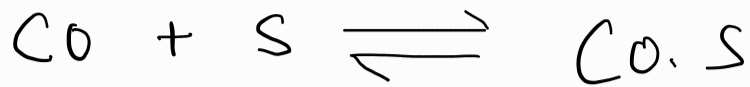
$\downarrow$
are observable

$$C_t = C_v + C_{A.S} \rightarrow \text{site balance}$$

$\hookrightarrow$ site where
adsorbed A is
present.



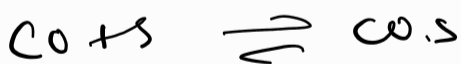
Non dissociative (Molecular adsorption)



Dissociative



depends on energy $\rightarrow$ if $\Delta H_{\text{ads}} > \text{Bond Energy}$.



$$r_{\text{ads}} = k_A p_{\text{CO}} C_V$$

$\hookrightarrow$ adsorption constant.

$$r_{\text{des}} = k_A C_{\text{CO} \cdot \text{S}}$$

Net rate of adsorption

$$r_{\text{Ad}} = r_{\text{ads}} - r_{\text{des}}$$

$$r_{\text{Ad}} = k_A p_{\text{CO}} C_V - k_{-A} C_{\text{CO} \cdot \text{S}}$$

$$r_{\text{Ad}} = k_A \left[p_{\text{CO}} C_V - \frac{C_{\text{CO} \cdot \text{S}}}{\frac{k_A}{k_{-A}}} \right]$$

$$\delta_{Ads} = K_A \left[p_{CO} C_V - \frac{C_{CO.s}}{K_A} \right]$$

where

$$K_A = \frac{k_A}{k_{-A}}$$

at equilibrium

$$\delta_{ads} = 0$$

$$p_{CO} \cdot C_V = \frac{C_{CO.s}}{K_A}$$

$$C_{CO.s} = K_A p_{CO} C_V$$

$$\Rightarrow C_V = \frac{C_{CO.s}}{K_A p_{CO}}$$

$$C_t = \frac{C_{co.s}}{K_A p_{co}} + C_{co.s}$$

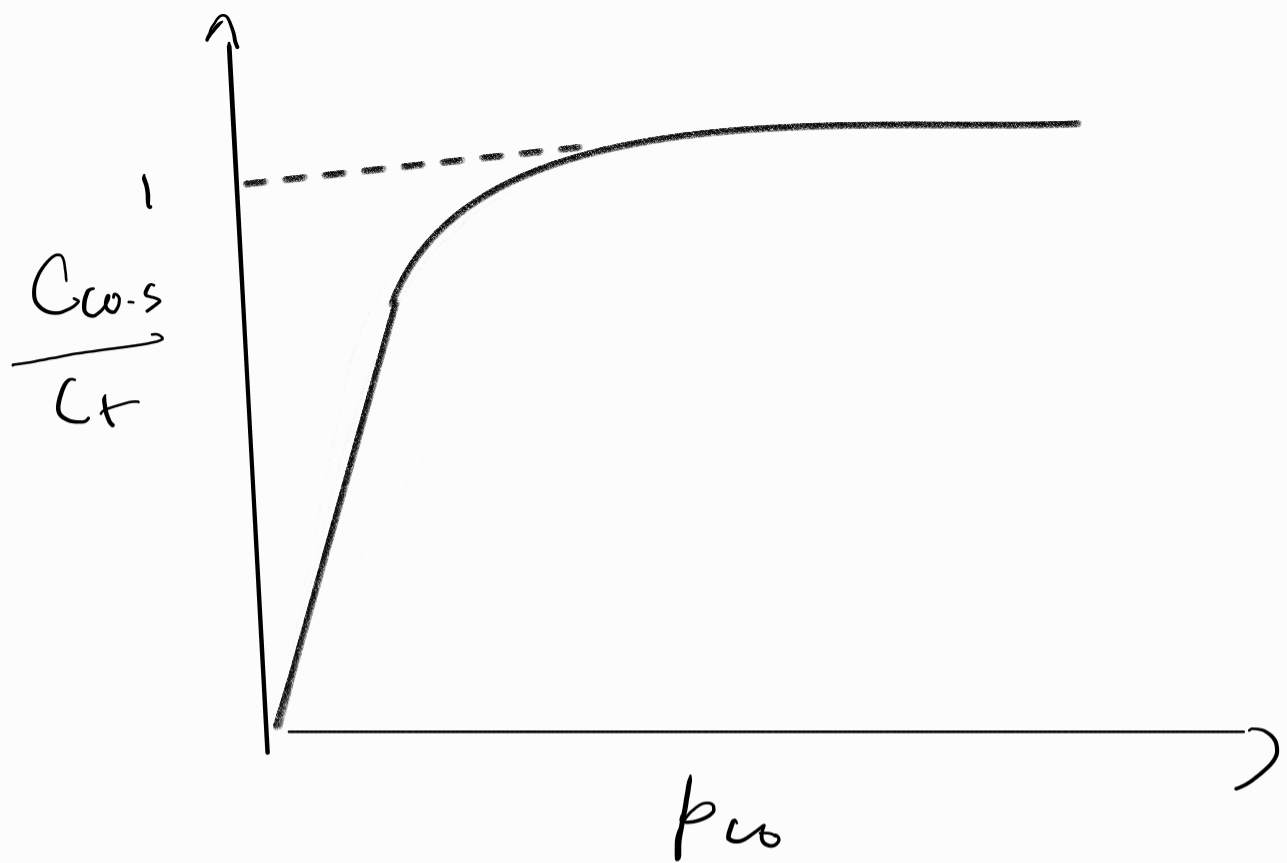
$$C_{co.s} \left[\frac{1 + K_A p_{co}}{K_A p_{co}} \right] = C_t$$

$$C_{co.s} = \frac{K_A p_{co} C_t}{1 + K_A p_{co}}$$

→ same eqn as
Langmuir isotherm

$$\frac{C_{co.s}}{C_t} = \theta = \frac{K_A p_{co}}{1 + K_A p_{co}}$$

→ can also linearize this eqn

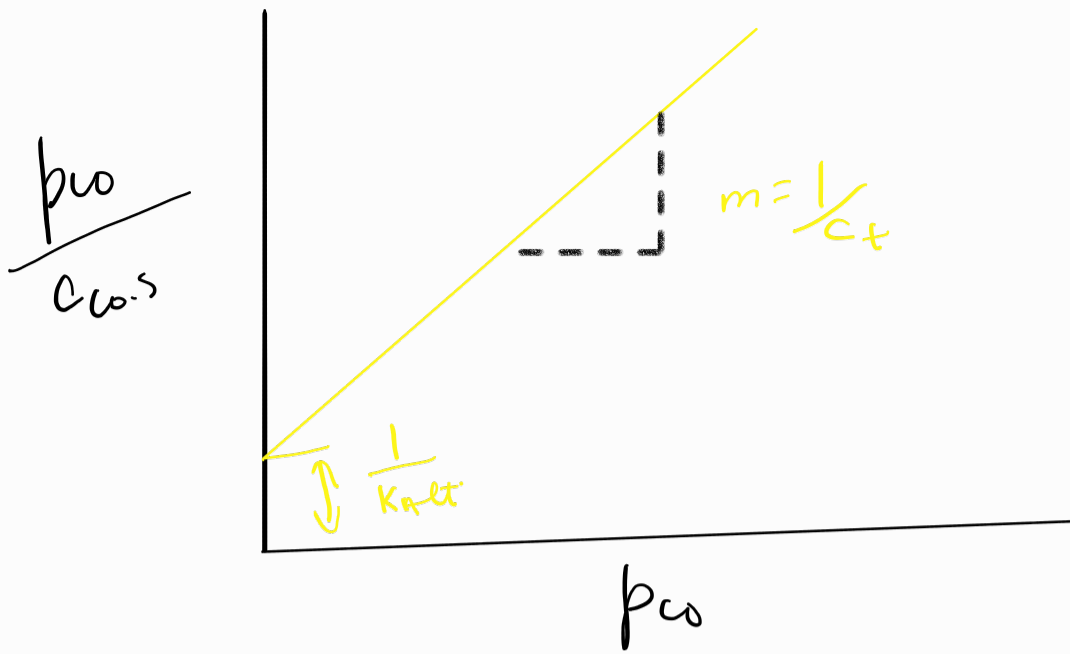


Reciprocal

$$\boxed{\frac{p_{w.s}}{C_{w.s}}} = \boxed{\frac{1}{K_A C_t}} + \frac{\textcircled{p_{w.s}}}{\textcircled{C_t}}$$

$\downarrow$ $\downarrow$ $\downarrow$
 y c $+ \quad m x$

x



Dissociative adsorption $\rightarrow$ dia, poly atomic molecules



$$\gamma_{ads} = k_A p_{CO} C_v^2$$

$$\gamma_{des} = k_{-A} C_{C.S} \cdot C_{O.S}$$

$$\gamma_{Ad} = k_A p_{CO} C_v^2 - k_{-A} C_{C.S} \cdot C_{O.S}$$

$$\gamma_{Ad} = K_A \left[p_{CO} C_v^2 - \frac{C_{c.s} C_{o.s}}{K_A} \right]$$

at equilibrium

$$\gamma_{Ad} = 0$$

→ assumption
 $C_{c.s} = C_{o.s}$

$$\Rightarrow p_{CO} C_v^2 = \frac{C_{c.s} \cdot C_{o.s}}{K_A}$$

$$p_{CO} \cdot C_v^2 = \frac{C_{c.s}^2}{K_A}$$

$$C_{c.s} = \sqrt{p_{CO} K_A} \cdot C_v$$

$$C_t = C_v + C_{c.s} + C_{o.s}$$

$$C_t = C_v + 2C_{c.s}$$

$$C_t = \frac{C_{c.s}}{\sqrt{p_{CO} K_A}} + 2C_{c.s}$$

$$C_t = C_{c.s} \left[2 + \frac{1}{\sqrt{p_{CO} K_A}} \right]$$

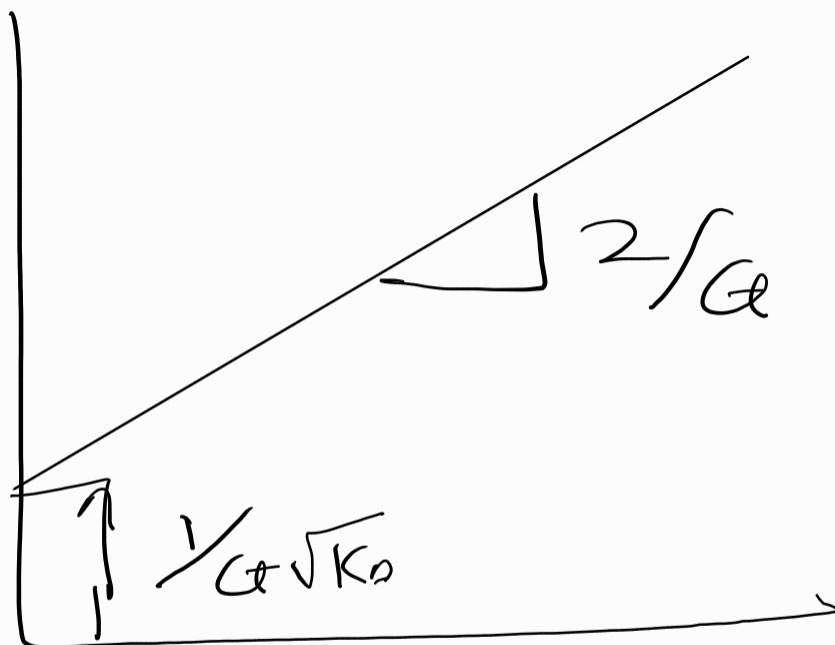
$$\frac{C_{c.s}}{C_t} = \frac{\sqrt{p_{CO} K_A}}{2\sqrt{p_{CO} K_A} + 1}$$

→ x

$$\frac{\sqrt{p_{CO}}}{C_{c.s}} = \frac{1}{C_t \sqrt{K_A}} + \frac{2\sqrt{p_{CO}}}{C_t}$$

y
C
m

$$\frac{\sqrt{b\omega}}{C_x \cdot S}$$



$$\sqrt{b\omega}$$

$$\frac{C_{co.7}}{C_e} = \frac{K_A p_D}{1 + K_A p_D}$$

$$\frac{C_{co.1}}{t} = \frac{\sqrt{K_A p_{co}}}{1 + 2\sqrt{K_A p_{co}}}$$

Non dissociative

Dissociative

Assumptions

- 1) Dynamic equilibrium
- 2) Elementary
- 3) No adsorbate - adsorbate interaction

(A)

(B)



$$\frac{C_{AS}}{C_e} = \frac{K_A p_A}{1 + K_A p_A + K_B p_B}$$

$$\frac{C_{B.S}}{C_t} = \frac{k_B p_B}{1 + k_A p_A + k_B p_B}$$



$$r_{ads} = k_{A1} p_A C_S - k_{A1}' C_{AS}$$



$$r_{ads} = k_{B1} p_B C_S - k_{B1}' p_B S$$

$$\left(C_{A1} C_S - \frac{1}{K_A} C_{AS} \right)$$

$$p_A C_S = \frac{1}{K_A} C_{AS}$$

$$\Rightarrow C_S = \frac{C_{AS}}{p_A K_A}$$

$$p_B C_S = \frac{1}{K_B} C_{BS}$$

$$C_T = C_S + C_{AS} + C_{BS}$$

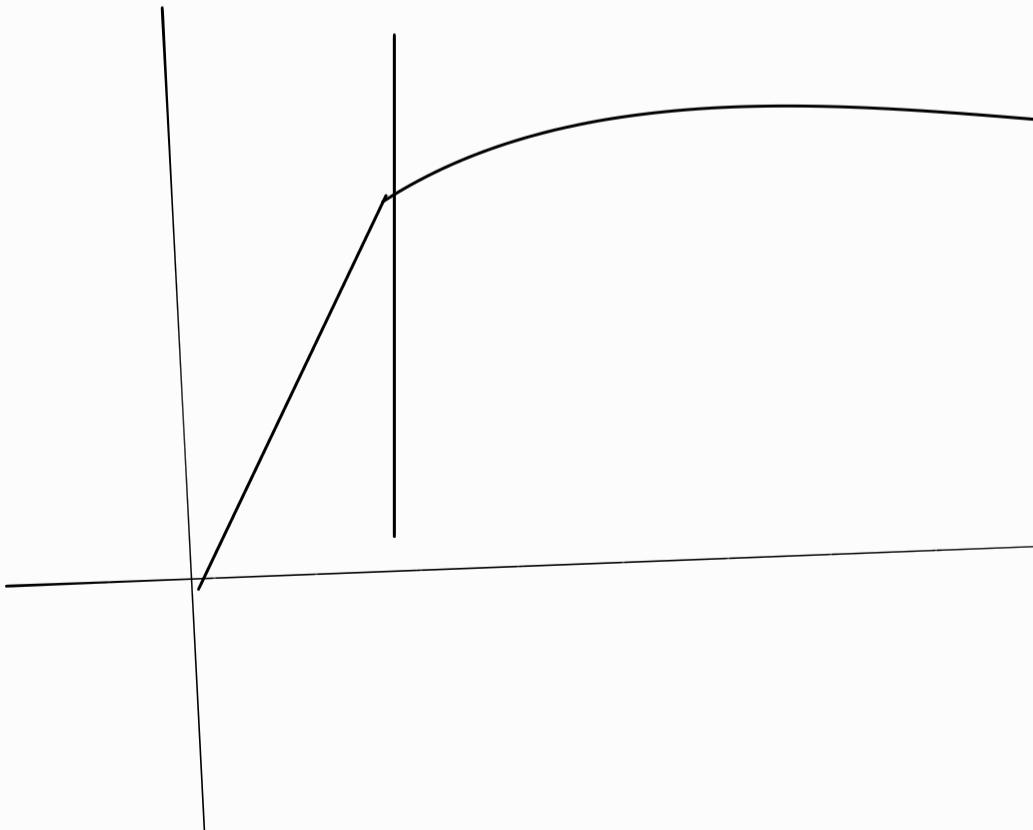
$$C_T = (1 + K_A p_A + K_B p_B) C_S$$

$$\frac{1}{1 + K_A p_A + K_B p_B} = \frac{C_S}{C_T}$$

$$C_S = \frac{C_{A.S}}{P_A K_D}$$

$$C_S = \frac{C_{B.S}}{P_B K_B}$$

$\Rightarrow$



Van't Hoff Equation

$$\frac{d \ln k}{dT} = \frac{\Delta H}{RT^2} \rightarrow \text{heat of adsorption}$$

$$\Rightarrow \frac{d \ln k}{d(1/T)} = \frac{-\Delta H}{R}$$

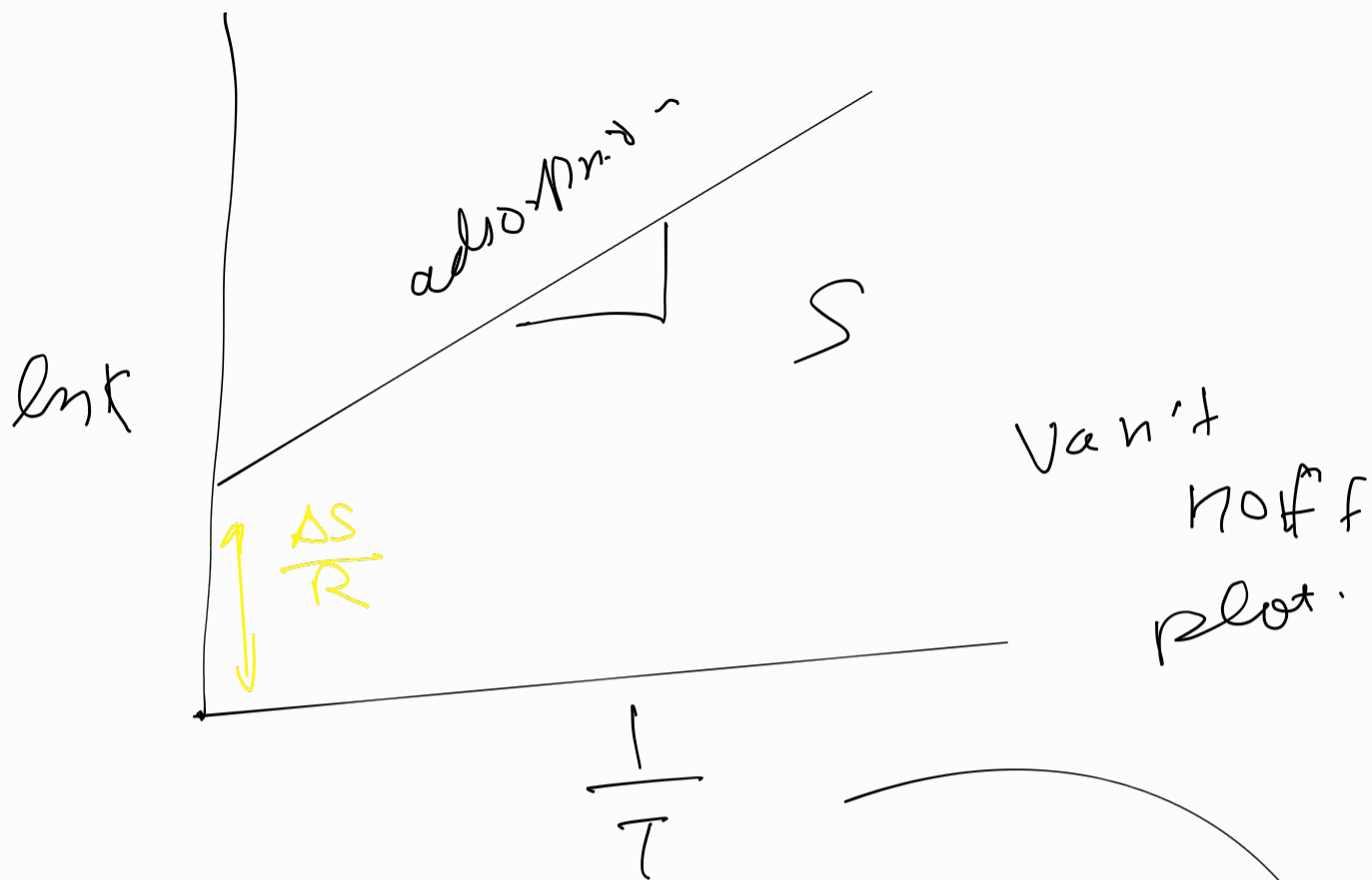
const 'T' & 'P'

$$\Delta G = \Delta H - T\Delta S$$

$$-RT \ln(K) = \Delta H - T\Delta S$$

$$\ln(k) = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

y x c



$$S = -\frac{\Delta H}{R}$$

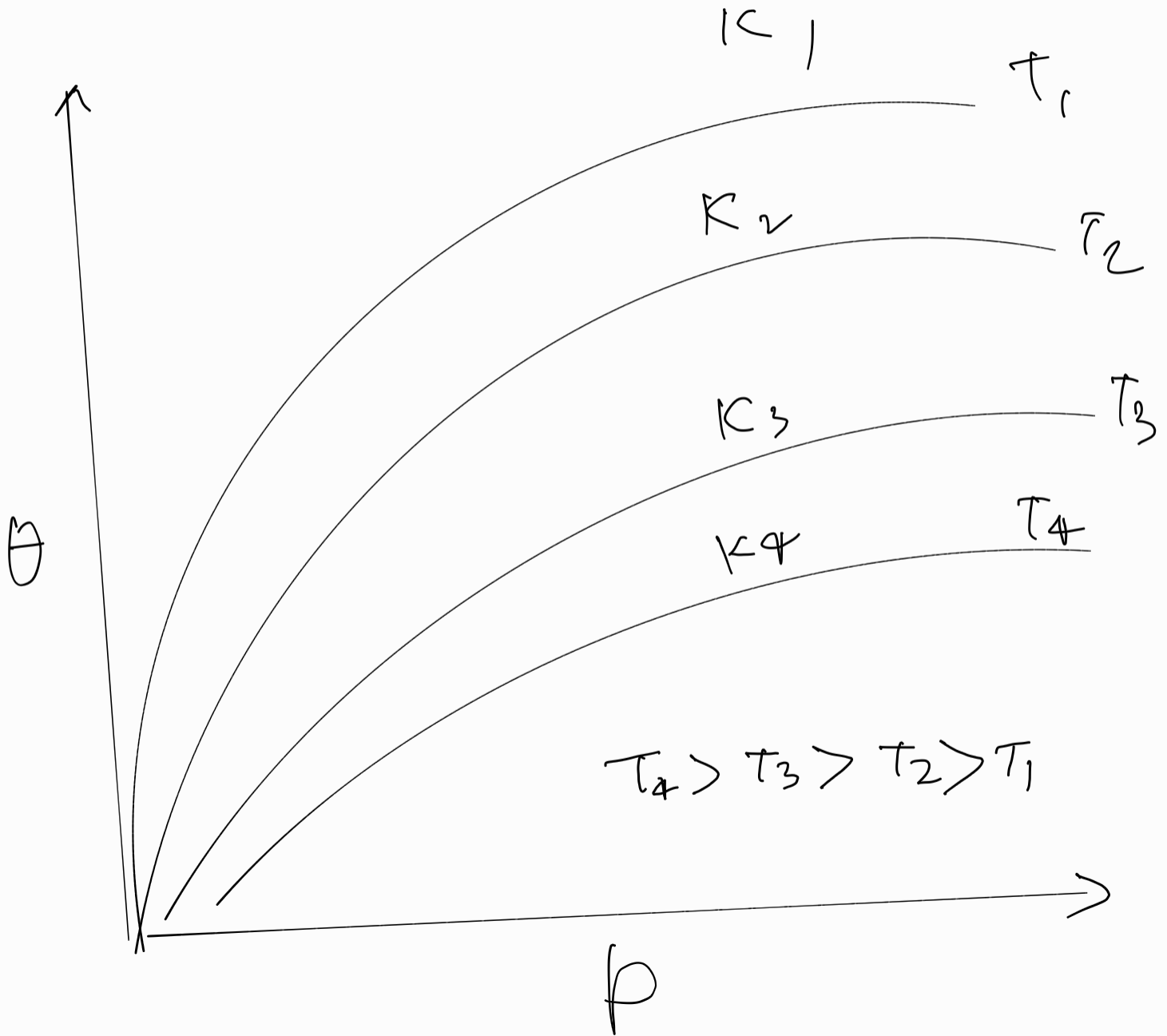
→ true for absorption

$$\Delta H = -S \times R$$

fitting

very wide range of T

$$\theta = \frac{kp}{1+kp}$$



Absorption isotherm

① Langmuir

② Freundlich

③ T₀H₀

④ BET

surface area

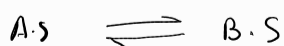
Total surface (support & active sites)



$$-r_A = k_A (p_A C_V - \frac{C_{A \cdot S}}{K_A})$$

$$-r_A = \frac{C_A}{\frac{1}{k} + \frac{1}{k_{ga}}}$$

Surface Reaction



single site

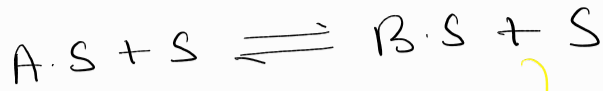
$$r_{Af} = k_s C_{A \cdot S}$$

$$r_{Ab} = k_{-s} C_{B \cdot S}$$

$$\begin{aligned} r_{As} &= k_s C_{A \cdot S} - k_{-s} C_{B \cdot S} \\ &= k_s \left[C_{A \cdot S} - \frac{C_{B \cdot S}}{K_s} \right] \end{aligned}$$

$$\left\{ K_s = \frac{k_s}{k_{-s}} \right\}$$

Dual Site

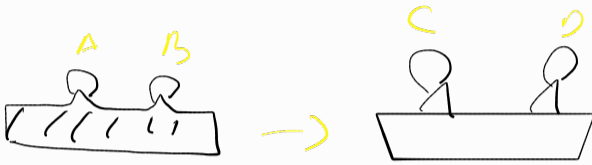


$$r_f = k_s C_{A \cdot S} C_S$$

$$r_b = k_{-s} C_{B \cdot S} C_S$$

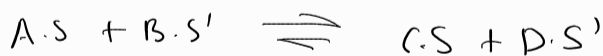
↪ rehydration
of alcohol

$$r_{AS} = k_s \left[C_{A \cdot S} C_V - \frac{C_{B \cdot S} C_V}{K_s} \right]$$



$$r_{AS} = k_f C_{A \cdot S} C_{B \cdot S}$$

$$r_{AS} = k_s \left[C_{A \cdot S} C_{B \cdot S} - \frac{C_{C \cdot S} C_{D \cdot S}}{K_s} \right]$$

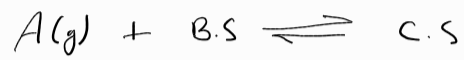


$$r_{AS} = k_s \left[C_{A \cdot S} C_{B \cdot S'} - \frac{C_{C \cdot S} C_{D \cdot S'}}{K_s} \right]$$

Langmuir - Hinshelwood Kinetics

Eley Riedel

A



$$-r_A = k_s p_A C_{B,s} \quad r_{A \rightarrow} = k_s C_{C,s}$$

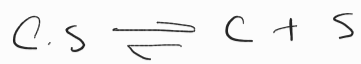
$$r_{A \rightarrow} = k_s p_A C_{B,s} - k_s C_{C,s}$$

$$r_{A \rightarrow} = k_s \left[p_A C_{B,s} - \frac{C_{C,s}}{K_s} \right]$$

$$\Delta G^\circ = RT \ln(K_s)$$

$$K_s = \frac{k_s}{k'_s}$$

Desorption



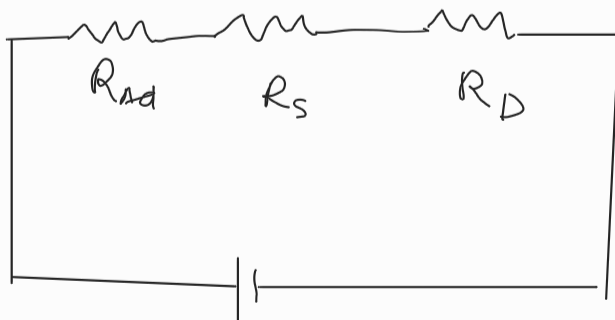
$$r_{des} = k_d C_{C \cdot S} - k_{-d} p_C C_S$$

$$r_D = k_D \left[C_{C \cdot S} - \frac{p_C C_V}{K_{DC}} \right]$$

Desorption
eqn const

$$K_{DC} = \frac{1}{K_C} \rightarrow \text{absorption eqn const.}$$

$$r_D = k_D \left[C_{C \cdot S} - K_C p_C C_V \right]$$



$$\Sigma R = 100 \Omega$$

$$I = \frac{V}{\Sigma R}$$

say

$$R_1 = 99$$

$$R_2 = R_3 = 0.5$$

$$r = \frac{\Delta c (\Delta b)}{1 + \sum k_i p_i (c_i)}$$

kinetic rate constant

$$r = \frac{k (\Delta c)}{1 + \sum k_i p_i}$$

driving force

Resistance

No of species getting absorbed

Langmuir-Hinshelwood Houghen Watson (LHHW) kinetics

$$-r_A = \frac{k (p_A - p_B)}{1 + K_A p_A + K_B p_B}$$

$k = e^{-E_A/RT}$

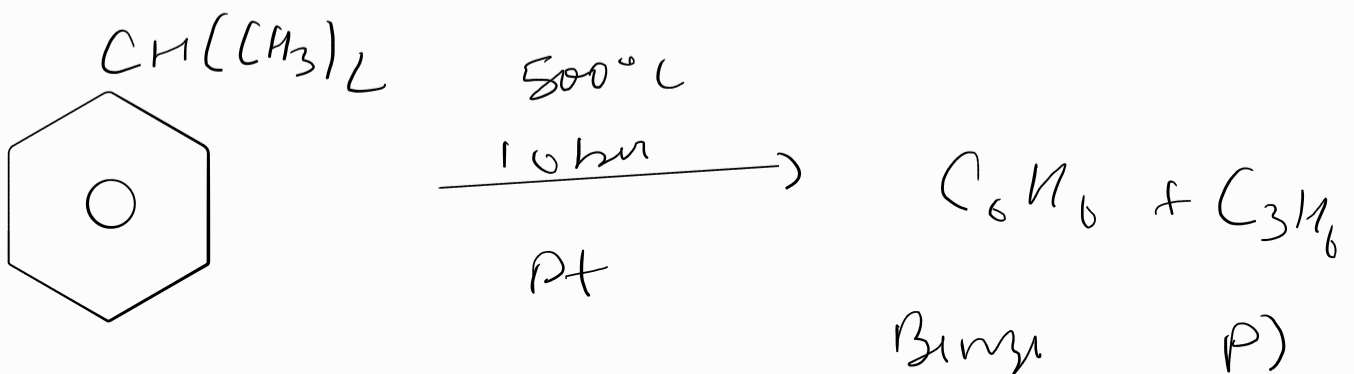
$f(C_A, T)$

$\Delta G = -RT \ln K_a$

Fogger $\Rightarrow$ ch-10

JM Smith

Synthesising rate Law



Common.

Q Derive a suitable rate law.

Adsorption: $\text{C} + \text{S} \rightleftharpoons \text{C}\cdot\text{S}$

Surface reaction: $\text{C}\cdot\text{S} \rightleftharpoons \text{B}\cdot\text{S} + \text{P}$

Desorption: $\text{B}\cdot\text{S} \rightleftharpoons \text{B} + \text{S}$

$$-r_c' = r_{AD} = r_s = r_D$$

Case-I I_s adsorption rate limiting?

$$r_{AD} = k_A \left[p_c C_v - \frac{C_{c,s}}{K_c} \right]$$

$$r_s = k_s \left[C_{c,s} - \frac{C_{B,s} p_p}{K_s} \right]$$

$$r_D = k_D \left[C_{B,s} - \frac{p_B C_v}{K_{DB}} \right]$$

$$= k_D \left[C_{B,s} - K_B p_B C_v \right]$$

$$K_B = \frac{1}{K_{DB}}$$

$$k_0 \ll k_s \sim k_0$$

$$\frac{\gamma_s}{k_s} \sim \frac{\gamma_D}{D} \sim 0$$

$$\frac{\gamma_s}{k_s} \sim 0 \Rightarrow C_{C.S} - \frac{C_{B.S} p_p}{k_s} \approx 0$$

$$C_{C.S} = \frac{C_{B.S} p_p}{k_s}$$

— (1)

$$\Rightarrow C_{B.S} = k_B p_B C_V$$

— (2)

$$\frac{\gamma_D}{k_D} \sim 0$$

$$\Rightarrow C_{B.S} = k_B p_D C_V$$

$$C_{s.s} = \frac{K_B p_B C_v p_P}{K_S}$$

$$-r_C = r_{AD} = k_A \left[p_C C_v - \frac{C_{c.s}}{K_C} \right]$$

→ (3)

Substitute $C_{c.s}$ from eq(2) to eq(3)

$$r_{AD} = k_A \left[p_C C_v - \frac{K_B p_B C_v p_P}{K_S} \right]$$

$$= k_A C_v \left[p_C - \frac{K_B p_B p_P}{K_S} \right]$$

Site Balance

$$C_T = C_{B.S} + C_{c.s} + C_v$$

$$= K_B p_B C_V + \frac{K_B p_B p_O C_V}{K_S} + C_V$$

$$= C_V \left[K_B p_B + \frac{K_B p_B p_O}{K_S} + 1 \right]$$

$$\Rightarrow C_V = \frac{C_t}{\left[K_B p_B + \frac{K_B p_O p_P}{K_S} + 1 \right]}$$

Substitute in eq(4)

$$-r_c = r_{Ap} = \frac{\overset{\text{Limit C}}{\underbrace{C_t k_A}} \left[\overset{\text{driving force}}{p_C - p_B p_O K_B / K_S} \right]}{\underbrace{\left[1 + K_B p_O + \frac{K_B p_B p_P}{K_S} \right]}_{\text{Resistance}}}$$

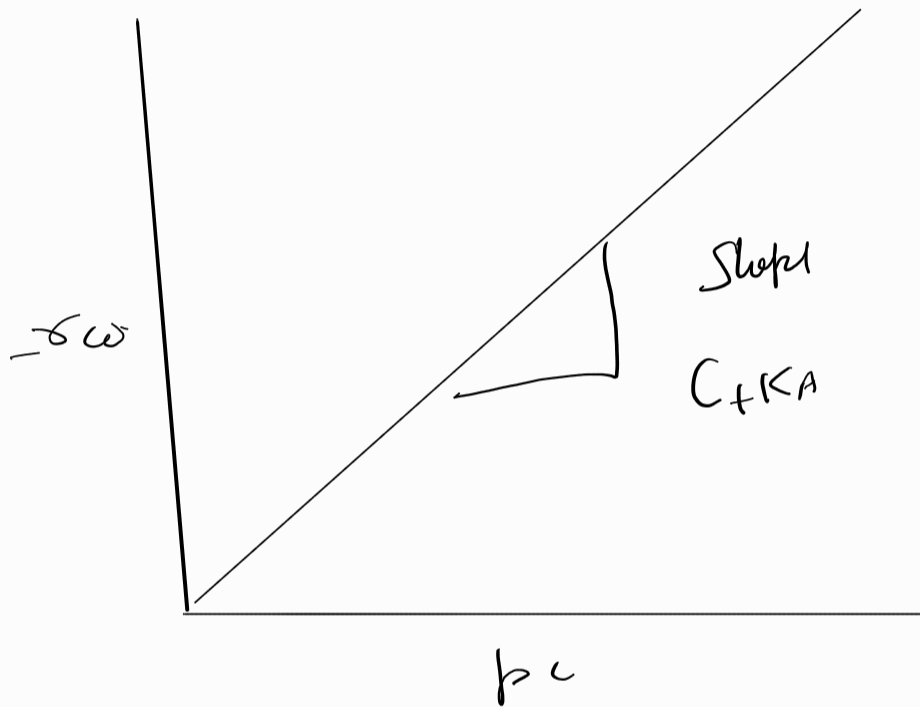
experimental

data fitting.

initially,

$$p_B \sim p_P \sim 0$$

$$\Rightarrow -r_{CO} = r_{AD} = C_t K_A p_C$$



Case-II

Surface reaction rate limiting

$$k_s \ll k_A \sim k_D$$

$$\frac{r_s}{k_s} \gg \underbrace{\frac{r_{AD}}{k_A} \sim \frac{r_D}{k_D}}_{> 0}$$

$$\delta_{AD} = k_A \left[p_C C_V - \frac{C_{C.S}}{K_C} \right]$$

$$\text{if } \frac{\delta_{AD}}{k_D} \sim 0$$

$$p_C C_V = \frac{C_{C.S}}{K_C}$$

$$C_{C.S} = K_C p_C C_V$$

$$\delta_D = k_D \left[C_{B.S} - K_B p_B C_V \right]$$

$$\text{if } \frac{\delta_D}{k_D} \sim 0$$

$$C_{B.S} = K_B p_B C_V$$

$$\left[C_c \cdot s - \frac{C_B \cdot s \cdot p_p}{k_s} \right]$$

$$\delta_s = k_s C_v \left[k_c p_c - \frac{K_B p_B p_p}{V_s} \right]$$

$$= k_s C_v k_c \left[p_c - \frac{K_B p_B p_p}{k_s k_c} \right]$$

$$K_p = \frac{K_s K_c}{K_B}$$

$$\delta_s = k_s C_v k_c \left[p_c - \frac{p_B p_p}{K_p} \right]$$

Site Bindu

$$C_T = C_{B.S} + (C_{C.S} + C_V)$$

$$C_T = [k_B p_B + k_C p_C + 1] C_V$$

$$\Rightarrow C_V = \frac{C_T}{[1 +]}$$

$-r_c$

$$= r_s = k_s K_C C_t [p_C - p_B p_P / K_P]$$

$$\frac{[1 + k_B p_B + k_C p_C]}{[1 + k_B p_B + k_C p_C]}$$

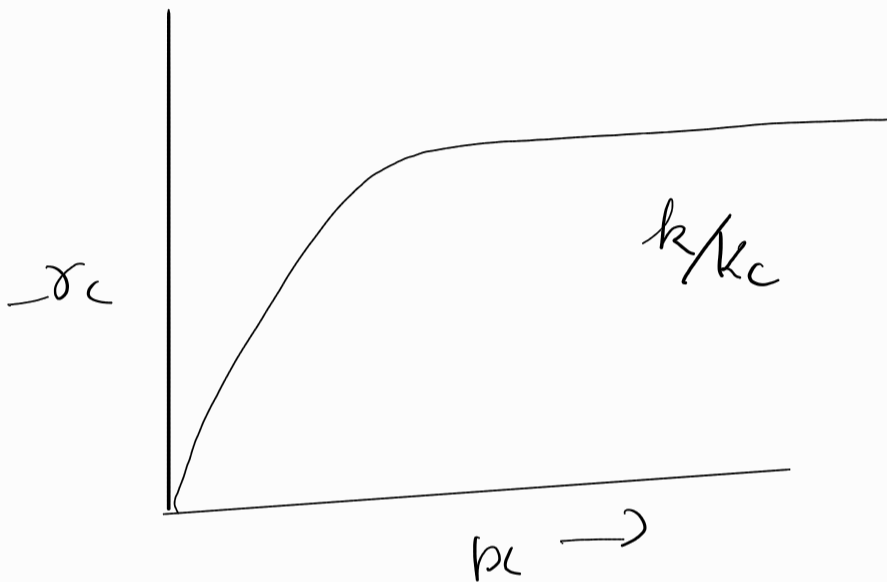
$$= k [p_C - p_B p_P / K_P] \frac{[1 + k_B p_B + k_C p_C]}{[1 + k_B p_B + k_C p_C]}$$

initially if $p_3 - p_0 \sim 0$

$$-\delta_{co} = \frac{k p_c}{1 + k_c p_c}$$

if $p_c \uparrow$

$$k_c p_c \gg 1$$



$$\delta_s = \frac{k p_c}{k_c p_c}$$

Case - III

In adsorption set
limit

$$k_D \ll k_A \sim k_S$$

$$\frac{\delta_D}{k_D} \gg \frac{\delta_{AD}}{k_A} \sim \frac{\delta_S}{k_S}$$

$$C_{C.S} = K_C p_C C_V$$

$$C_{B.S} = \frac{K_C p_C K_S C_V}{p_P}$$

$$-r_C = r_D = k_D [C_{B.S} - K_B p_B C_V]$$

$$= C_V k_D \left[\frac{K_C p_C K_S}{p_P} - K_B p_B \right]$$

Site - balance

$$C_T = C_{B.S} + C_{C.S} + C_V$$

$$C_T = \left[\frac{K_c \mu_c K_s}{p_o} + K_c \mu_c + 1 \right] C_v$$

$$\Rightarrow C_v = \frac{C_t}{\left[\frac{K_c \mu_c v_s}{p_p} + K_c \mu_c + 1 \right]}$$

$$-r_c = r_D = \frac{k_D C_t \left[K_c \mu_c K_s / p_p - K_B \mu_B \right]}{\left[1 + \frac{K_c \mu_c K_s}{p_B} + K_c \mu_c \right]}$$

$$= \frac{k_D K_c K_s / p_p C_t \left[\mu_c - \frac{p_B \mu_B}{K_p} \right]}{\left[p_p + K_c K_s \mu_c + K_c \mu_c p_p \right]}$$

$$\sigma_D = \overbrace{k_D K_C K_S}^{K_{\text{overall}}} C_t \left[p_c - \frac{p_B p_p}{K_p} \right]$$

(driving force)

$$\left[p_p + K_C K_S p_c + K_C \cdot p_c p_p \right]$$

(system)

L H H W.

$$-r_A = \frac{k (p_A - p_B / K_p)}{1 + K_A p_A + K_B p_B}$$

Date: 31/8/26

silica



Run	$\sigma_T \times 10^0$	Toluene	Hydrogen	Methane	Benzene
Set A					
1	71.0	1	1	1	0
2	71.3	1	1	4	0

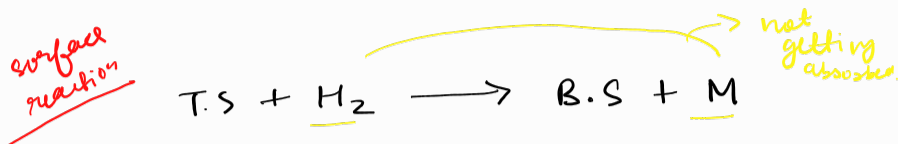
Set B					
3	41.6	1	1	0	1
4	19.7	1	1	0	4

Set C

Eley Rideal mechanism

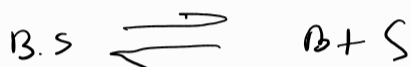
$$\sigma = \frac{k \cdot k_d P_A \cdot P_B}{1 + K_A \cdot P_A}$$

LHHW Kinetics



$$r_{T.S} = k' C_{T.S} p_{H_2}$$

desorption



$$r_{des} = k_d \left[C_{B.S} - \frac{p_B C_v}{K_{DB}} \right]$$

$$r_{des} = k_d \left[C_{B.S} - p_B C_v K_B \right]$$

$$-r_T' = \frac{k p_{H_2} p_B}{1 + K_B p_B + K_T p_T}$$

① surface reaction is irreversible

② H, M are not absorbed.

We know

$$\frac{1}{K_{DB}} = K_B$$

if surface reaction is the RDS

$$k \ll k_A \sim k_d$$

Pseudo Steady State Hypothesis (PSSH)

$$C_{T.S} \sim C_{B.S} \sim 0$$

$$r_{T.S} \sim r_{B.S} \sim 0$$

$$r_{ad} \sim r_{des} \sim 0$$

$$C_{T.S} = K_T p_T C_V$$

$$\sigma_d \sim 0$$

$$C_{B.S} = K_B p_B C_V$$

$$\sigma_s = K' C_{T.S} p_{H_2}$$

$$\sigma_s = K' K_T p_T C_V p_{H_2}$$

$$C_T = C_{T.S} + C_{B.S} + C_V$$

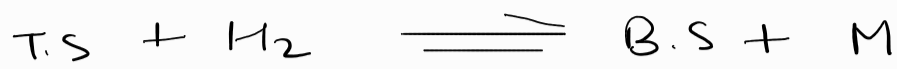
$$= K_T p_T C_V + K_B p_B C_V + C_V$$

$$C_V = \frac{C_T}{1 + K_B p_B + K_T p_T}$$

$$-\sigma_T' = \sigma_S = \frac{\overset{\textcolor{yellow}{k}}{\textcolor{red}{\curvearrowright}} k_C k_T p_T p_{H_2}}{1 + k_T p_T + k_B p_B}$$

$$-\sigma_T' = \sigma_S = \frac{k p_T p_{H_2}}{1 + k_T p_T + k_B p_B}$$

Q Derive an expression for $-r_T'$, assuming surface reaction is reversible & it is still vds



$$r_s = k_s C_{T.S} p_{H_2} - k_{-s} C_{B.S} p_M$$

$$= k_s \left[C_{T.S} p_{H_2} - \frac{C_{B.S} p_M}{K_s} \right]$$

$$= k_s \left[K_T p_T C_V p_{H_2} - \frac{K_B p_B C_V p_M}{K_s} \right]$$

$$C_V = \frac{C - 1}{1 + K_T p_T + K_B p_B}$$

$$\frac{k_s C_T \left(K_T p_T p_{H_2} + \frac{K_B p_B p_M}{K_s} \right)}{1 + K_T p_T + K_B p_B}$$

$$-r_T' = r_S = k_S C_T \left[K_T p_T p_{H_2} + K_B p_B p_M \right] \frac{K_S = 1}{K_{SO}}$$

$$1 + K_T p_T + K_B p_B$$

Now that we have got the rate expression we have to do non-linear curve fitting.

to find the parameters (K_T, K_B, k)

Now that we have got the rate

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